



# Intelligent crude oil price probability forecasting: Deep learning models and industry applications

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## ABSTRACT

The crude oil price has been subject to periodic fluctuations because of seasonal changes in industrial demand and supply, weather, natural disasters and global political unrest. An accurate forecast of crude oil prices is of utmost importance for decision makers and industry players in the energy sector. Despite this, the volatility of crude oil prices contributes to the uncertainty of the energy industry, which was particularly challenging following the recent global spread of the COVID-19 epidemic and the Russia–Ukraine conflict. This paper proposes a hybrid deep learning (DL) modelling framework to deal with the volatility of crude oil prices, applying ensemble empirical mode decomposition (EEMD), convolutional neural network (CNN) and bidirectional long short-term memory (BiLSTM) integrated with quantile regression (QR); named EEMD-CNN-BiLSTM-QR. Two real-world datasets on crude oil prices from the West Texas Intermediate and Brent Crude Oil markets were employed to validate the EEMD-CNN-BiLSTM-QR hybrid modelling framework. Given that the probability density forecast can capture uncertainty, an in-depth analysis was carried out and prediction accuracy calculated. The findings of this study demonstrate that the proposed EEMD-CNN-BiLSTM-QR DL modelling framework, which uses the probability density forecast method, is superior to other tested models in terms of its ability to forecast crude oil prices. The novelty of this study stems mainly from its use of QR, which allows for the description of the conditional distribution of predicted variables and the extraction of more uncertain information for probability density forecasts.

## 1. Introduction

Crude oil is a vital global product, and price fluctuations significantly influence global financial, economic and political stability (Baumeister and Kilian, 2015, Dai et al., 2024, Dong et al., 2024). For example, the 2008 Global Financial Crisis caused significant fluctuations in the pricing of Brent Crude Oil. The price dropped from US\$143.95 per barrel on 3 July to US\$33.73 on 26 December of that year. Moreover, the emergence of unexpected global calamities, notably the unprecedented COVID-19 pandemic, amplifies the challenges of accurately forecasting the volatility of crude oil prices (Huang and Deng, 2021). The remarkable event of West Texas Intermediate (WTI) crude oil prices reaching an unprecedented low of −US\$37.63 per barrel on 20 April 2020 was primarily caused by the lag in crude oil production compared with a significant decline in crude oil demand. More recently, the conflict between

Russia and Ukraine has been seen as a critical factor causing the price of crude oil to fluctuate dramatically. Since Russia is among the world's top oil-producing nations and energy suppliers, it has already seen the Brent oil price jump to over US\$100 per barrel, the highest it has been since 2014 (Liadze et al., 2022). Price fluctuations are more complicated and non-linear in nature because of an increasingly broad spectrum of risk variables (Li et al., 2021a). Consequently, accurate forecasts of crude oil prices play a significant role in the energy industry. However, because of the complexity of the energy system, forecasting crude oil prices is challenging. Therefore, achieving accurate predictions of future crude oil prices is of significant interest to policymakers, governments, commercial entities, decision makers, market leaders and financial investors (Abdollahi, 2020, Xing and Zhang, 2022, Shang et al., 2024).

Extensive research has been conducted to explore the volatility of crude oil prices, and various methods for predicting crude oil prices have

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been proposed (Abdollahi, 2020, Li et al., 2021a, Drachal, 2021, Monge and Gil-Alana, 2021, Alqahtani et al., 2021). For example, a method known as generalised autoregressive conditional heteroscedasticity (GARCH) was used by Alqahtani et al. (2021) to investigate whether the COVID-19 pandemic led to financial impacts caused by jump processes in crude oil prices. However, it is difficult for GARCH, as a traditional econometrics method, to effectively capture the non-linearity and volatility of crude oil prices. To overcome these shortcomings, the time-frequency decomposition technique has been applied before establishing the model. For example, Abdollahi (2020) provided a hybrid model based on time series decomposition using GARCH to improve the accuracy of the prediction of crude oil prices. Monge and Gil-Alana (2021) applied wavelet analysis in conjunction with fractional integration, and the fractional cointegration vector autoregressive (FCVAR) technique, to investigate how regional crude oil production divergence affects WTI crude oil prices. Moreover, based on the variational mode decomposition (VMD) technique, Li et al. (2021a) developed a novel multi-scale forecasting model to enhance the accuracy of complex crude oil price time series. However, it is worth noting that the preceding research concentrated on the point forecast rather than the probability density forecast. Although substantial progress has been made in the accuracy of predicting crude oil prices, the uncertainty of crude oil prices cannot be captured using point forecasts. A probability density forecast can capture and interpret the uncertainty of information about crude oil prices. From a statistical perspective, it is not difficult to find that uncertain global events lead to abnormal fluctuations in crude oil prices (Li et al., 2019). In the point forecast, the modeller uses historical data to estimate specific data following the ordinary least squares (OLS) method and tends to the mean or median of the specific data. Regression techniques based on the OLS method focus only on variation in the mean, which produces biased estimates (Dawar et al., 2021). That is, the specific result obtained deviates from the expected value during the period of abnormal fluctuation. However, probability density forecasting refers to estimating the corresponding probability of all possible future results of an undetermined variable. Probability prediction can provide more information for decision makers than can point prediction. For this reason, it is reasonable to argue that the point forecast has limited application. Hence, the undetermined probability of forecasting crude oil prices should be explicitly considered to improve and more meaningfully evaluate the performance of prediction techniques.

Despite the influence of various stochastic factors in the energy market, probability density forecasting can effectively quantify the uncertainty associated with crude oil price prediction by acquiring the quantile of crude oil prices at different time points (Costa et al., 2021). Quantile regression (QR) is a technique for implementing probability density forecasts that is extensively used with machine learning (ML) in the energy market. Deep neural networks (NNs) are widely applied in complex non-linear time series modelling to avoid the local optima and over-fitting of traditional ML. For example, the convolutional neural network (CNN), one of the most influential deep learning (DL) models, was employed to extract features of raw time series data through a convolution kernel to capture the undetermined, complicated non-linear features of the volatility of crude oil prices, mainly considering the underlying spatial feature (Chena et al., 2017, Dong et al., 2024). Furthermore, a bidirectional long short-term memory (BiLSTM) NN was applied to model the future volatility of crude oil price series, mainly considering the temporal feature (Kakade et al., 2022). Therefore, to improve the robustness of modelling, a QR-based DL ensemble model is used in this study to produce probability density forecasts of crude oil prices by effectively exploiting temporal-spatial features.

In summary, this study aims to propose a novel hybrid model of probability density forecasting to quantify the uncertainty of crude oil prices using DL NNs. According to He et al. (2022), there has recently been a shift from focusing solely on optimising individual prediction techniques as the only means to increase load forecasting accuracy. The proposed hybrid modelling framework may be broken down into three

distinct stages. First, the time-frequency technique decomposes the initially received raw signal to generate several intrinsic mode functions and a residual component. Second, DL NNs are employed to extract temporal-spatial features of crude oil prices. Finally, probability density forecasting is applied with the QR function as the outcome. Two datasets were used to validate the performance of the proposed modelling framework: WTI and Brent Crude Oil, which are extracted from the United States of America (USA) oil fields and North Sea oil fields, respectively. The empirical results are measured from the perspective of point forecasting and probability density forecasting, respectively. Afterward, an in-depth analysis is conducted to assess the effectiveness of the prediction. The following is a detailed illustration of the contributions made in this study:

1. A probability density forecast modelling framework that incorporates QR with time-frequency techniques and DL NNs is proposed in this study, which can provide a comprehensive evaluation of future trends in crude oil prices. Compared with the case of point forecasts, probability density forecasts of crude oil prices can provide point prediction results and interval predictions at each time scale and capture a complete probability distribution of crude oil prices by evaluating the likelihood of future trends. Research on the probability density forecast of crude oil prices based on DL is scarce.
2. An ensemble NN structure of probability density forecast based on DL is programmed. The proposed framework uses CNN-BiLSTM to extract the relevant temporal-spatial features of influential variables. Each component extracted by ensemble empirical mode decomposition (EEMD) is reconstructed into  $d$  dimensional explanatory variables. In the ensemble model of CNN-BiLSTM-QR, the CNN model is employed to extract spatial features, while the BiLSTM NN is used to mine temporal features. QR, the last layer of network structure, is applied to establish a regression prediction model under specific conditional quantiles. Each NN in the ensemble model fulfills a distinct function, broadening the application of probabilistic approaches based on DL for forecasting crude oil prices.
3. Using the EEMD-CNN-BiLSTM-QR model enables energy industries and traders to enhance the precision of their trading methods, hence optimising their trading and purchasing decisions in the crude oil market. Moreover, the proposed model can seamlessly incorporate and evaluate diverse real-time data, resulting in more precise and prompt predictions. This technique mitigates ambiguity for energy corporations, allowing them to make more informed decisions about production, investment and resource distribution. Two real-world daily and weekly datasets from Brent Crude Oil and WTI were used to validate the performance of the proposed intelligent crude oil price probability forecasting modelling framework.

The remainder of this paper is structured as follows. Section 2 provides a brief literature review on crude oil price forecasting techniques. Section 3 introduces the fundamental methodologies and the proposed hybrid modelling framework. Then, the experimental results, including datasets, evaluation metrics and experimental techniques, are discussed in detail in Section 4. Section 5 analyses the experimental results. Section 6 discusses the proposed modelling framework. Finally, the conclusion of this study is provided in Section 7.

## 2. Literature review

This section concisely reviews relevant literature on probability density forecasting in the energy market and highlights the novelty of probability density forecasting in crude oil pricing research. Crude oil—a non-renewable energy source—is one of the most significant commodities traded on the global market and plays a crucial function in society and socioeconomic, geopolitical and industrial arenas (Li et al., 2021a). However, two fundamental factors may affect crude oil prices at varying intervals: price fluctuations and an imbalance between demand

**Table 1**  
Studies on probability density forecast in the energy market.

| Authors                     | Modelling framework                                                                                       | Field of study          | Time scale             | Limitation                                                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Chai et al. (2018)          | Bayesian                                                                                                  | Crude oil price         | Monthly                | Data was limited for modelling                                                                                               |
| Bai et al. (2022)           | Continuous ranked probability score (CRPS) combination and equal-weighted combination (ensemble approach) | Crude oil price         | Monthly                | An error in parameter estimation                                                                                             |
| Costa et al. (2021)         | Several ML modelling techniques                                                                           | Oil price               | Monthly, quarterly     | The difference between performance based on the interval score and that on the log predictive density score is not clear-cut |
| He et al., (2019b)          | Support vector quantile regression                                                                        | Wind and solar power    | Daily                  | The feature of the pattern is manually selected                                                                              |
| Zhang et al. (2021b)        | NN-based QR integrating with Gaussian kernel function                                                     | Day-head load           | Daily                  | Classical gradient descent algorithm brings out local optimum and over-fitting                                               |
| Yang et al. (2012)          | Autoregressive conditional interval models with exogenous explanatory interval variables                  | Crude oil price         | Daily                  | An energy market with stochasticity makes it challenging to meet the condition of the prior distribution hypothesis          |
| Wang et al. (2020)          | AGWOCS                                                                                                    | Crude oil price         | Daily, weekly          | Support vector regression (SVR) makes it difficult to conduct feature selection automatically during pattern recognition     |
| Huang et al. (2021)         | Boosting and MA                                                                                           | Crude oil price         | Daily                  | Classical gradient descent algorithm brings out local optimum and over-fitting                                               |
| Sun et al. (2021)           | A quantile autoregression model                                                                           | Crude oil price         | Daily                  | The high non-linearity and non-stationarity of crude oil prices make it challenging to support the proposed framework        |
| HE et al., (2019a)          | Least absolute shrinkage and selection operator with quantile regression neural network (LASSO-QRNN)      | Electricity consumption | Yearly                 | Classical gradient descent algorithm brings out local optimum and over-fitting                                               |
| Ding et al. (2021)          | C-MIDAS-X with mean squared forecast error (MSFE) combination                                             | Steam coal price        | Weekly                 | There is a call for applying other probability density functions for forecasting research                                    |
| He et al. (2020)            | VMD with quantile regression forest (QRF) and kernel density estimation (KDE)                             | Load                    | Daily                  | The paper considered only the temporal features and ignored the spatial features of energy                                   |
| Ding et al. (2022)          | EEMD-MIDAS- SVR                                                                                           | Natural gas             | Daily                  | SVR makes it difficult to conduct feature selection automatically during pattern recognition                                 |
| Kim et al. (2020)           | Using a ML technique based on Bayesian                                                                    | Power                   | Daily                  | Respondent selection bias of survey data                                                                                     |
| He and Zhang (2020)         | QRNN with Epanechnikov kernel function                                                                    | Wind power              | Hourly                 | Classical gradient descent algorithm brings out local optimum and over-fitting                                               |
| Guo et al. (2018)           | A DL based on the QR and KDE model                                                                        | Electricity consumption | Weekly                 | The paper considered only the temporal features and ignored the spatial features of energy                                   |
| Wang et al. (2019)          | QR-based using random forest regression, gradient boosting decision tree, SVR and KDE                     | Electrical load         | Monthly                | SVR makes it difficult to conduct feature selection automatically during pattern recognition                                 |
| Uniejewski and Weron (2021) | LASSO with quantile regression averaging                                                                  | Electricity price       | Daily                  | SVR makes it difficult to conduct feature selection automatically during pattern recognition                                 |
| He and Wang (2021)          | EEMD-LASSO-QRNN                                                                                           | Wind power              | Daily                  | Classical gradient descent algorithm brings out local optimum and over-fitting                                               |
| Yang et al. (2018)          | Gaussian process quantile regression is a type of Bayesian non-parametric method                          | Power load              | Hourly                 | Heavy computational cost                                                                                                     |
| Zhang and Liu (2024)        | ICEEMDAN-FE-FIG-ARIMA-IESN                                                                                | Crude oil price         | Daily, weekly, monthly | The paper considered only the temporal features and ignored the spatial features of crude oil prices.                        |

and supply (Zhao et al., 2017). Since crude oil is a global energy resource, there is a constant and strong need to investigate and develop more accurate methods of predicting crude oil prices.

In the energy market, one of the most challenging tasks has been to provide an estimate of the degree of price uncertainty regarding crude oil that is sufficiently accurate. Although probability density forecasting has the potential to quantify the uncertainty that is associated with the volatilities of crude oil prices, it is clear that there is relatively little research on the probability density forecast of crude oil prices (Chai et al., 2018, Yang et al., 2012, Wang et al., 2020, Huang et al., 2021, Sun et al., 2021). For example, Xing and Zhang (2022) used shrinkage methods with different penalties and losses to forecast crude oil prices. They revealed that forecasts utilising a non-convex penalty and Huber loss outperformed various shrinkage predictions using alternative models. Wang et al. (2020) proposed an ensemble probabilistic prediction approach based on the adaptive cuckoo search algorithm (AGWOCS) for modelling uncertainty in daily and weekly Brent Crude Oil prices. However, the AGWOCS-based modelling framework is employed to resolve the issue of probabilistic prediction for crude oil prices. The shortcomings of convergence speed and local optima of AGWOCS cannot be avoided. Therefore, Huang et al. (2021) developed a reliable model by introducing a novel two-stage approach that utilised boosting and model averaging (MA) to forecast the interval values of crude oil prices. The method is designed to capture the uncertain features of movements in the market. However, boosting is sensitive to

abnormal samples, which may obtain higher weights during iteration, affecting the accuracy of the final prediction. Moreover, Sun et al. (2021) proposed a new interval prediction model integrating improved VMD and particle swarm optimisation (PSO) with bootstrap sampling for online real-time forecasting of crude oil prices. They also developed a quantile autoregression model to analyse the return and fluctuation of crude oil futures based on the price analysis of WTI. Recently, Zhang and Liu (2024) proposed an improved hybrid interval prediction model of crude oil prices combining improved complete EEMD with adaptive noise (ICEEMDAN) and fuzzy entropy (FE), fuzzy information granulation (FIG), improved echo state network (IESN), and an autoregressive integrated moving average model (ARIMA). However, a review of a sample of the relevant literature reveals little research on probability density forecasting of crude oil prices.

The methods described in the literature for probability density forecasts concerning the energy market provide valuable suggestions for probability density forecasts of crude oil prices. The following section provides several examples of the probability density function for energy industries. For example, utilising the semiparametric variation technique, Datta et al. (2017) examined the suitability of the distribution generated from various probability density function (PDF) estimating techniques and established a set of comprehensive outputs for the future oil price. However, the energy market's stochasticity makes it challenging to support the hypothesis of trade-off between pricing errors and PDF smoothness. Thus, to better understand the stochastic analysis,

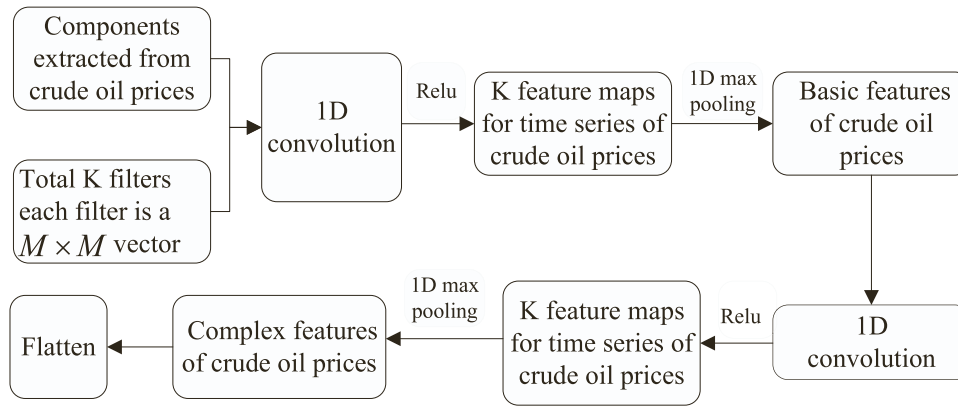


Fig. 1. The network structure in the 1D-CNN.

Maulion et al. (2022) proposed a framework for the mean square deviation in which the PDF demonstrated a periodic modulating pattern, exposing a non-Markovian stochastic process that functioned as comprehensive information in forecasting crude oil prices. However, the probability density prediction model has been applied in numerous energy research domains such as wind and solar power (He et al., 2019b; He and Wang, 2021), electricity load (Wang et al., 2019, Zhang et al., 2021b, Xu et al., 2021, Xu et al., 2019), and steam coal prices (Ding et al., 2021) and has achieved superior prediction accuracy.

Several studies have used a wide variety of cutting-edge modelling approaches based on artificial intelligence (AI) and sophisticated statistical methods to forecast the price of crude oil. For example, Ali Salamai (2023) acquired temporal patterns from input modes using DL by stacking several long short-term memory (LSTM) networks with BiLSTM. A hybrid short-term forecasting system was proposed by Hou et al. (2022) to predict the crude oil price, based on identifying outliers and an enhanced multi-objective water cycle algorithm. Based on a novel decomposition ensemble (i.e., VMD and FE; Guo et al., 2022), and EEMD and multi-layer regression (MLR; Jiang et al., 2022) strategy, a multi-perspective crude oil price forecasting model was suggested by Li et al. (2022a), who suggested and validated an enhanced secondary decomposition-based hybrid model, which provides new knowledge for investigating crude oil price forecasting. Moreover, an improved slope-based method utilising empirical mode decomposition (EMD) and a feed-forward neural network approach were proposed by Fang et al. (2022) to decompose and predict crude oil prices. In addition, a substantial number of significant crude oil forecasting models, including multi-stage (Li et al., 2021a) and hybrid models (Li et al., 2021b, Yang et al., 2021), have recently been suggested. Although research has shown that the crude oil forecasting methods have a significant level of prediction accuracy, the previously proposed frameworks do have a few drawbacks. For example, the parameters of many algorithms are found by trial and error. An additional shortcoming of the previous studies is that the point forecasting of crude oil prices may ignore fluctuation signals. Jiang et al. (2022) identified the possibility of achieving a higher degree of predicted accuracy of crude oil price by estimating intervals or through probability forecasting. Considering the limitations identified in earlier research, the current study proposes a novel hybrid framework named EEMD-CNN-BiLSTM-QR to predict the crude oil price in the context of a new decomposition ensemble framework. An overview of 22 studies on probability density forecasts for the energy market is provided in Table 1.

The following is a summary of the primary evidence presented in Table 1. The number of studies on probability density forecasts for crude oil prices is relatively low. Among the sample of studies in Table 1, only nine relate to crude oil prices and employ point estimator technique, such as Bayesian and bootstrap to deal with the interval prediction. The need for prediction interval (PI) techniques to adhere to the prior

distribution hypothesis is the method's most significant shortcoming because the substantial non-linearity and non-stationarity of crude oil prices make this challenging to achieve.

In addition, daily datasets were employed for most of the studies. Because of the significant effect of stochastic elements on short-term forecasting, daily and weekly crude oil prices of the WTI and Brent markets are used in the current study for probability density forecasts. Only one study in Table 1 used DL to make probability density forecasts. Despite the development of classical NNs in the energy market, these networks have not been able to learn how to deal with complicated time series. The current study combines CNN and BiLSTM into a network structure with QR to obtain the conditional quantile of crude oil prices at each time point. Finally, mode decomposition techniques have been integrated for the energy market to improve the accuracy of probability prediction in recent years. EEMD is combined in the modelling framework for the probability density forecast of crude oil prices.

### 3. Methodology

#### 3.1. Ensemble empirical mode decomposition

EEMD is a non-linear and self-adaptive time-frequency analysis algorithm developed by Wu and Huang (2009). It adds Gaussian white noise to a raw time series to resolve the inherent drawback of mode mixing based on the classical EMD. EEMD can effectively pull out the intrinsic characteristics of a complex time series to help modellers make better predictions (He and Wang, 2021, Ding et al., 2022).

The fundamental concept behind EEMD is quite similar to that for EMD, which first computes the envelope via spline interpolation and then determines the meaning of the envelope. Through a set of decompositions, the crude oil price time series  $X = \{x_1, x_2, \dots, x_T\}$  (where  $T$  is the time-length of samples) can be extracted step by step to generate a finite number of intrinsic mode functions (*imfs*) components ( $n \text{ imf}(t)$ ) from high to low frequency and a residue component  $r_n(t)$ . The components of decomposition obtained are shown in Eq. (1):

$$x(t) = \sum_{i=1}^n \text{imf}_i(t) + r_n(t), t = 1, \dots, T \quad (1)$$

where  $n$  is the number of *imfs* extracted and the residue component  $r_n(t)$  indicates the trend of crude oil prices. These *imfs* hold significant physical interpretations, aiding in the comprehension of underlying mechanisms driving price fluctuations.

#### 3.2. Convolutional neural network

Studies have indicated that CNN can effectively extract the non-linear spatial inherent relationship features from an input time series, improving prediction performance (Yao et al., 2021, Chen et al., 2021,



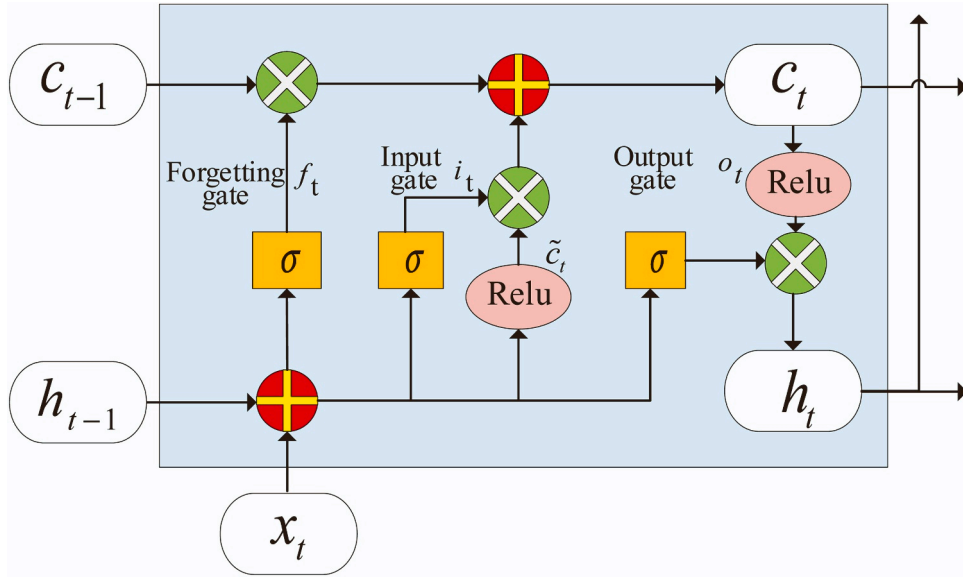


Fig. 2. The structure of the memory block.

Ren et al., 2022, Li et al., 2022b, He et al., 2021). The CNN, inspired by visual neuroscience, is composed a convolution layer and a pooling layer. The convolution layer can maintain the spatial continuity of the time series data and extract the local features of raw signals, and the pooling layer can adopt max pooling or mean-pooling to reduce the dimensions of the middle (hidden) layer and the computation of the following layers. Furthermore, because of local receptive fields and the shared weight of the convolution kernel, the complexity of the NN structure and the number of weight parameters are reduced significantly, simplifying the computational complexity of CNN (Li et al., 2022b).

As crude oil price data are an example of a typical non-linear time series, this study uses one-dimensional CNN (1D-CNN) to capture the structural information relevant to the current behaviour of crude oil prices. The network structure of the 1D-CNN consists of eight layers, as illustrated in Fig. 1. These layers are as follows: an input layer, two one-dimensional convolutional layers, two one-dimensional pooling layers, a flattened layer and two activation function layers. The convolution and pooling layers are connected alternately in the framework. The first layer of convolution and pooling will learn the basic features of crude oil prices, and the second layer will allow the model to learn more complex features. It is noted that there are three primary methods for convolution operation: full convolution, valid convolution and same convolution. To avoid the number of dimensions increasing through full convolution and a loss of information caused by valid convolution, the same mode is selected to perform the convolution. The convolution operation is defined as in Eq. (2):

$$y_{ij}^{(k)} = \phi \left( \sum_{p=1}^M \sum_{q=1}^M \omega_{pq}^{(k)} x_{i+p-1, j+q-1}^{(k)} + b^{(k)} \right), k = 1, \dots, K \quad (2)$$

where  $y_{ij}^{(k)}$  is the output vector of convolution operation in the  $i$ th row and  $j$ th column of the  $k$ th filter,  $\phi$  indicates the activation function,  $M$  is the size of the filter;  $\omega_{pq}^{(k)}$  is the weight in the  $p$ th row and  $q$ th column of the  $k$ th filter;  $x_{i+p-1, j+q-1}^{(k)}$  is the input vector after padding;  $b^{(k)}$  is the bias of the  $k$ th filter and  $K$  indicates the number of the convolution kernel. As the price of crude oil is a non-linear time series, the rectified linear unit (Relu) is employed as the activation function of the convolution operation.

Furthermore, max pooling, one of the most commonly used pooling methods, is employed to reduce the dimensions of feature maps and prevent over-fitting. It is important to note that the feature

representation extracted from the pooling layer is the main aim of our work to improve performance.

The components extracted by EEMD are fed into the input layer of the network structure. Then, they are convoluted and pooled by 1D-CNN to generate a new sequence of feature representations, which will be concatenated as the input of the network structure of BiLSTM.

### 3.3. Bidirectional long short-term memory

The BiLSTM method was proposed by Graves and Schmidhuber (2005) as an extension of the LSTM NN, which can effectively solve the problem of gradient vanishing or explosion of the recurrent neural network in the case of learning the long-term time series dependencies, through introduction of a gate control unit in the memory block. The structure of the memory block is shown in Fig. 2. Three gates control the memory and output: the forgetting gate, the input gate and the output gate. Compared with LSTM, the BiLSTM NN can better capture long-term information dependencies from two-directional memory propagating forwards and backwards.

In this study, we employ two BiLSTM layers to capture the non-linear temporal long-term inherent relationship. As shown in Fig. 3, each layer of the BiLSTM network includes two sub-networks: a forward LSTM NN and a backward LSTM NN.  $X = \{x_1^1, x_2^1, \dots, x_T^1\}$  is fed into the BiLSTM structure, and then  $H = \{h_1, h_2, \dots, h_T\}$  is generated as the prediction after modelling.  $\vec{h}_t^i$  and  $\overleftarrow{h}_t^i$  denote the output of the forward LSTM NN and the output of the backward LSTM NN at time step  $t$  of the  $i$ th BiLSTM layer, respectively. The final output  $H^i$  is concatenated with  $\vec{h}^i = \{\vec{h}_1^i, \dots, \vec{h}_T^i\}$  and  $\overleftarrow{h}^i = \{\overleftarrow{h}_1^i, \dots, \overleftarrow{h}_T^i\}$ . The formula for the computational process is defined as follows:

$$\vec{h}_t^i = LSTM(x_t^i, \vec{h}_{t-1}^i), t = 1, \dots, T \quad (3)$$

$$\overleftarrow{h}_t^i = LSTM(x_t^i, \overleftarrow{h}_{t-1}^i), t = 1, \dots, T \quad (4)$$

$$H^i = \vec{h}^i + \overleftarrow{h}^i \quad (5)$$

where  $x_t^i$  denotes the input at time  $t$  of the  $i$ th BiLSTM layer and  $T$  is the time series length.

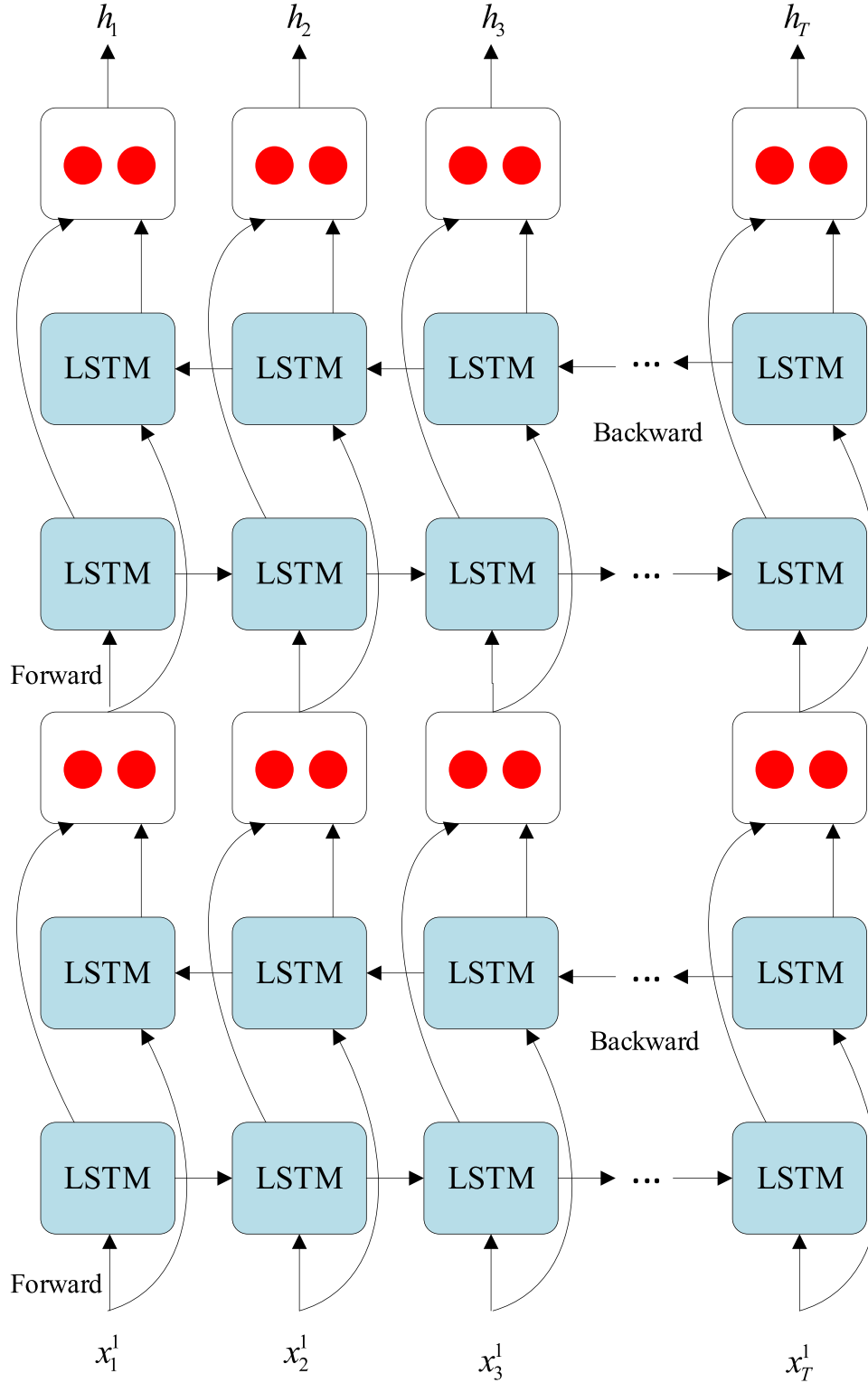


Fig. 3. The structure of the BiLSTM network.

#### 3.4. Quantile regression

The last layer of the network structure is the QR layer, which establishes a regression model prediction under specific conditional quantiles. As indicated at the beginning of this paper, the price of crude oil is influenced by many undetermined factors, and the determined

forecasting model can offer only limited or inadequate deviation information because it is based on a departure from the hypothesis that a normal distribution is applicable. Fortunately, [Koenker and Gilbert Bassett \(1978\)](#) proposed the technique of QR, which captures the relationship between explanatory variables and conditional quantiles of the response variables without requiring a hypothesis or any prior

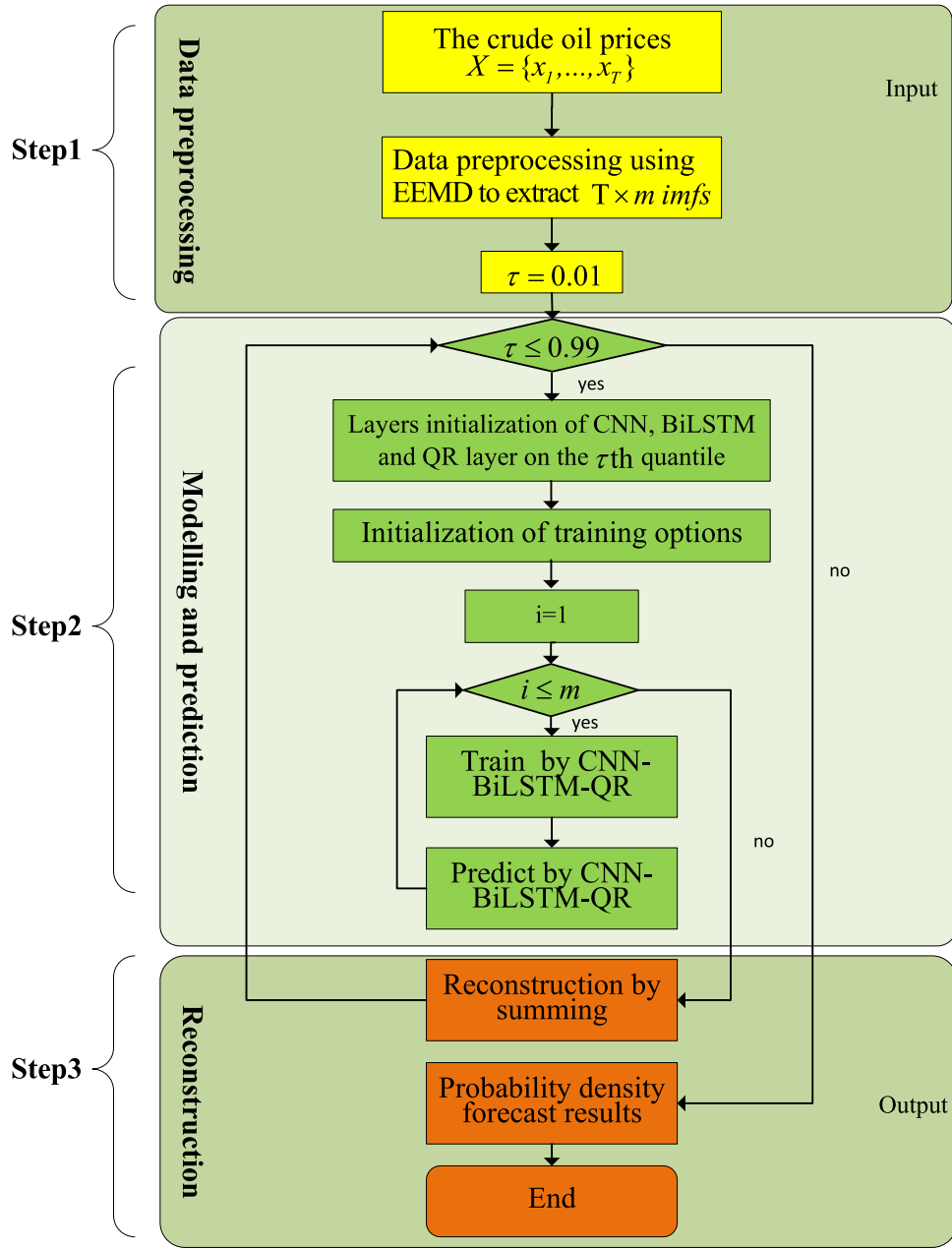


Fig. 4. The modelling framework of EEMD-CNN-BiLSTM-QR.

information. Compared with traditional regression models, QR can robustly and precisely characterise the conditional distribution of explanatory variables under different quantiles than classical regression models. A series of conditional quantiles for each time step obtained is utilised to calculate the probability density distribution of response variables.

According to the principle of the QR algorithm,  $F(y) = P\{Y \leq y\}$  denotes the distribution function of the response variables  $Y = \{y_1, y_2, \dots, y_N\}$ . The  $d$ -dimensional explanatory variables  $X = \{X_1, X_2, \dots, X_d\}$  are shown as follows:

$$\begin{Bmatrix} X_1, X_2, \dots, X_d \end{Bmatrix} = \begin{Bmatrix} x_{11}, & x_{12}, & \dots, & x_{1d} \\ x_{21}, & x_{22}, & \dots, & x_{2d} \\ \dots & \dots & \dots & \dots \\ x_{N1}, & x_{N2}, & \dots, & x_{Nd} \end{Bmatrix} \quad (6)$$

The  $\tau$ th conditional quantile of the response variables  $Y$  is defined by

Eq. (7):

$$Q_Y(\tau|X) = \beta_0(\tau) + \beta_1(\tau)X_1 + \dots + \beta_d(\tau)X_d = \beta(\tau)X^T \quad (7)$$

where  $\tau \in (0, 1)$  and  $\beta(\tau) = \{\beta_0(\tau), \beta_1(\tau), \dots, \beta_d(\tau)\}$  is the regression coefficient under the  $\tau$ th quantile.

Unlike traditional regression models, the QR algorithm uses an estimator of the weight least absolute deviation to evaluate the parameters of the coefficient  $\beta(\tau)$ , as shown in Eq. (8):

$$\hat{\beta}_\tau = \underset{\beta \in R^{d+1}}{\operatorname{argmin}} \sum_{i=1}^N \rho_\tau(y_i - x_i \beta(\tau)) \quad (8)$$

where  $x_i = [x_{i1}, x_{i2}, \dots, x_{id}] (i = 1, \dots, N)$  denotes the  $i$ th row vector of the explanatory variables, and  $\rho_\tau$  is a check function on the  $\tau$ th conditional quantile based on the difference between  $y_i$  and  $x_i \beta(\tau)$ , formula as in Eq. (9):

$$\rho_{\tau} = \begin{cases} \tau, & y_i - x_i\beta(\tau) \geq 0 \\ \tau - 1, & y_i - x_i\beta(\tau) \leq 0 \end{cases} \quad (9)$$

Thus, the evaluation of the response variables  $Y$  on the  $\tau$ th conditional quantile is acquired and expressed as follows:

$$\hat{Q}_{\tau}(X) = \hat{\beta}(\tau)X^T \quad (10)$$

With the prediction of the response variables among all the quantiles (i.e.,  $\tau \in (0.01, \dots, 0.99)$ ), short-term probabilistic density forecasting is generated at each time step.

### 3.5. The proposed EEMD-CNN-BiLSTM-QR modelling framework

The proposed model is a novel hybrid approach using DL NNs for probability density forecasts for crude oil prices. CNN, a most influential DL technique, has superior function approximation owing to its high-speed parallel computing capabilities, robustness and adaptive learning pattern. Although traditional NNs can certainly adapt to the non-linearity and non-stationarity of crude oil price time series, the classical gradient descent algorithm has a non-convex issue, which causes local optimum and over-fitting easily. In this study, CNN is employed to extract features of crude oil price time series data through a convolution kernel to capture the undetermined and complicated non-linear features of the volatility of crude oil prices. The primary purpose of CNN is to extract the spatial features of historical data. It has been shown that the CNN technique can successfully extract hidden spatial features from static data using convolutional computations (Hu et al., 2022). The BiLSTM NN, another influential DL method, is applied to predict the future volatility of long-term crude oil price time series. It should be emphasised that during training, an LSTM NN with a memory module overcomes the shortcomings of gradient vanishing or gradient exploding. However, one-directional LSTM only uses backward propagation to input previous information and ignores subsequent information in the historical data. Therefore, using BiLSTM as an extension of LSTM can significantly enhance the accuracy of time series forecasting (Ghasemlounia et al., 2021, Peng et al., 2021). In addition, like the modelling approaches used by AI, the optimum preprocessing strategy for crude oil prices is a crucial factor that significantly contributes to forecasting accuracy. Thus, an EEMD-based hybrid model, following the divide and conquer strategy, has been conducted for forecasting crude oil prices (Abdollahi, 2020, Wu et al., 2019).

Furthermore, undetermined forecasting is presented with a choice of quantitative assessment in uncertain prediction, another significant research topic covered in this study. Based on the findings of a comprehensive analysis of the relevant literature, undetermined predictions were made in this study using one of three methods: PIs, quantiles and PDF (Zhang et al., 2021b, Bracale et al., 2020). PI techniques such as Bayesian (Nandram, 1995, Lai et al., 2022, Chai et al., 2018), bootstrap (Ji et al., 2019) and lower upper bound estimation (LUBE; Taormina and Chau, 2015) are used to directly estimate the lower and upper bounds of the PIs, which are based on the prior distribution hypothesis. Nevertheless, even guided by a hypothetical distribution, the prediction of PIs often diverges from the predicted value in complex stochastic contexts. In contrast to PIs, quantiles are used as the output of undetermined prediction, which quantitatively prevents the hypothesis of distribution when evaluating uncertain prediction. An examination of the relevant literature from the past decades reveals that QR is used extensively in conjunction with ML in the energy market. QR is a method that generates relevant features as inputs and utilises those features to determine the upper and lower limits of PIs. As an example, He et al. (2019b) employed QR to successfully combine with support vector regression (SVR) for wind and solar power probability density forecasting. To determine the probability density distribution based on the quantile, kernel density estimation (KDE) may be used. This technique is performed by transforming the components obtained by QR into the appropriate form for KDE (Zhang et al., 2021b). Unlike the first two

**Table 2**

Period and size of experimental datasets (daily).

| Sample data | The period of the time series | Sample size | Training set | Testing set |
|-------------|-------------------------------|-------------|--------------|-------------|
| Brent       | 1 June 1987–25 June 2021      | 8649        | 5767         | 2882        |
| WTI         | 6 January 1986–30 July 2021   | 8964        | 5978         | 2986        |

methods, PDF can provide point prediction results and interval predictions for crude oil prices at each time scale and capture a complete probability distribution of crude oil prices by evaluating the likelihood of future trends (He et al., 2019b).

Based on the above techniques, this study proposes a hybrid prediction model that functions within the QR-based modelling framework and utilises EEMD to assemble CNN and BiLSTM for the probability density forecast of crude oil prices. The modelling framework of the proposed EEMD-CNN-BiLSTM-QR is shown in Fig. 4, which includes three steps as follows:

**Step 1: Data preprocessing.** Rooted in EMD principles, EEMD ensures alignment with physical interpretations, offering enhanced interpretability and noise resilience. Through the sifting process of EEMD, the non-stationary and non-linear sequences of crude oil prices are adaptively decomposed into  $m$  *imfs* with tangible interpretations, aiding understanding of the inherent mechanisms of price fluctuations.

**Step 2: Modelling and prediction.** An ensemble NN structure of a probability density forecast based on DL is programmed. Each component extracted from Step 1 is reconstructed to  $d$ -dimensional explanatory variables fed into the CNN layer of the network structure, which can capture a new sequence of spatial features by convolution and pooling operations. Then, the new sequence extracted using CNN will be concatenated as the input of the BiLSTM layer of the network, which can better capture long-term dependency information from two-directional memory propagating forwards and backwards. The CNN-BiLSTM layer in the network structure is introduced to model and forecast by capturing both temporal and spatial features. QR, the last layer of the network structure, is applied to establish a regression prediction model under specific conditional quantiles. The forecasting result determined for each component by the CNN-BiLSTM layer is applied as input into the QR layer, which effectively captures the relationship between explanatory variables and conditional quantiles of the response variables.

**Step 3: Reconstruction.** The components' predictions are reconstructed by summing under each quantile to generate an ensemble prediction for the original time series. Finally, the PDF results are computed as the final output.

## 4. Research design

### 4.1. Datasets

To validate the proposed EEMD-CNN-BiLSTM-QR forecasting model, two real-world datasets<sup>1</sup> were considered as experimental samples: the WTI market from the USA and the Brent Crude Oil market from the North Sea. Both daily and weekly data are considered, focusing on the significant influence that stochastic factors will have on short-term forecasting. Besides, since WTI and Brent Crude Oil are the two most famous benchmarks in international crude oil markets and play an important role in the pricing mechanism of all other global oils (Abdollahi, 2020, Xiong et al., 2013, Zhang et al., 2021a), the two benchmark datasets were selected to provide empirical evidence for

<sup>1</sup> Data are available from the Energy information Administration website (<http://www.eia.gov>)



**Table 3**

Period and size of experimental datasets (weekly).

| Sample data | The period of the time series | Sample size | Training set | Testing set |
|-------------|-------------------------------|-------------|--------------|-------------|
| Brent       | 5 June 1987–25 June 2021      | 1778        | 1185         | 593         |
| WTI         | 10 January 1986–30 July 2021  | 1856        | 1237         | 619         |

probability density forecasting in this study. Table 2 and Table 3 present the daily and weekly experimental datasets, respectively.

Each dataset for the crude oil price time series is split into the training and testing sets. The first two-thirds are for training, and the rest is for testing. As shown in Table 2, a daily series of Brent Crude Oil prices from 1 June 1987–25 June 2021 (8649 daily observations) is employed for PDF. The first 5767 observations, from 1 June 1987–5 February 2010, are used as a training set, and the last 2882 observations, from 8 February 2010–25 June 2021, are adopted as a testing set. A daily series of WTI crude oil prices from 6 January 1986–30 July 2021 (8964 daily observations) is used. The first 5978 observations, from 6 January 1986–14 September 2009, are used as a training set, and the last 2988 observations, from 15 September 2009–30 July 2021, are adopted as a testing set. Similarly, a weekly series of Brent Crude Oil prices from 5 June 1987–25 June 2021 (1778 weekly observations) is employed. The first 1185 observations, from 5 June 1987–12 February 2010, are used as a training set, and the last 593 observations, from 19 February 2010–25 June 2021, are adopted as a testing set. A weekly series of WTI crude oil prices from 10 January 1986–30 July 2021 (1856 weekly observations) is used. The first 1237 observations, from 10 January 1986–18 September 2009, are used as a training set, and the last 619 observations, from 25 September 2009–30 July 2021, are used as a testing set.

As the time series of crude oil prices are non-stationary, two datasets are standardised into the range of [0,1] before being fed into the proposed modelling framework (Chena et al., 2017, Drachal, 2021). All the models considered for comparison and then contrasted against one another are put into practice for density forecasting of the crude oil prices in the Brent and WTI markets. The evaluation results are generated for the testing sets. The prediction of the testing sets is used to evaluate the performance of each model in terms of the accuracy of determined forecasting and probability density forecast, respectively. This is done by comparing the results to the predictions of the testing sets.

#### 4.2. Evaluation metrics

To verify the performance of the proposed EEMD-CNN-BiLSTM-QR modelling framework for the PDF of crude oil prices, no single accuracy metric can capture the characteristics of the statistic and error distribution for the probability density forecast. Therefore, we compare the two types of error feature in determined prediction and undetermined prediction.

##### 4.2.1. Determined prediction evaluation metrics

Determined prediction performance is measured using two statistical evaluation metrics: mean absolute percentage error (MAPE) and root mean square error (RMSE). MAPE is a standard measure employed in many forecasting fields (He and Zhang, 2020, Ren et al., 2022, Guan et al., 2021), as the formula in Eq. (11). The MAPE indicates the relative degree of error between the prediction and observation. RMSE is chosen as the absolute measure of error, another primary evaluation metric (He et al., 2020, Chen et al., 2021, Pedro et al., 2018), as defined in Eq. (12):

$$MAPE = \frac{1}{T} \sum_{t=1}^T \left| \frac{y_t - \hat{y}_t}{y_t} \right| \times 100 \quad (11)$$

$$RMSE = \sqrt{\frac{\sum_{t=1}^T (y_t - \hat{y}_t)^2}{T}} \quad (12)$$

where  $T$  is the number of the testing set, and  $y_t$  and  $\hat{y}_t$  denote the observed value and the predicted value, respectively. It is clear that the smaller the values of MAPE and RMSE, the better the model performance.

##### 4.2.2. Undetermined prediction evaluation metrics

To better demonstrate the superiority of EEMD-CNN-BiLSTM-QR proposed in this paper, undetermined prediction performance is measured in terms of four evaluation metrics: the Pinball loss function, prediction interval coverage probability (PICP), prediction interval normalised averaged width (PINAW), and continuously ranked probability score (CRPS). The Pinball loss is a quantile score used to measure the accuracy of the quantile forecast (Guan et al., 2021, Xu et al., 2019), calculated using Eq. (13).

$$Pinball\left(\tau, y_t, \hat{y}_{\tau,t}\right) = \begin{cases} (1-\tau)(\hat{y}_{\tau,t} - y_t), & y_t < \hat{y}_{\tau,t} \\ \tau(y_t - \hat{y}_{\tau,t}), & y_t \geq \hat{y}_{\tau,t} \end{cases} \quad (13)$$

where  $\tau = 0.01, 0.2, \dots, 0.99$ , and  $y_t$  and  $\hat{y}_{\tau,t}$  denote the observation at time step  $t$  and the prediction under the  $\tau$ th quantile at time step  $t$ , respectively. The lower the Pinball score, the better the QR forecasting under the  $\tau$ th quantile.

Moreover, the reliability, sharpness and resolution of undetermined forecasting are commonly considered. PICP denotes the probability of the observation value falling between the upper and lower bounds of the PIs, which can be used to evaluate the reliability of PIs quantitatively (Pedro et al., 2018, He and Wang, 2021). The formula for PICP is shown in Eq. (14). The larger the value of PICP, the higher the reliability of the PIs. PINAW represents the width of PIs (Pedro et al., 2018, He and Wang, 2021), which describes the sharpness of undetermined forecasting, as expressed in Eq. (15). The smaller the value of PINAW, the better the sharpness of PIs. Probability prediction aims to improve the sharpness of prediction as much as possible under the premise of meeting PICP. However, there are constraints between PICP and PINAW (Zhang et al., 2021b). CRPS can be competent for the assessment of reliability as well as sharpness and can evaluate the resolution of performance in probability forecasting comprehensively (Pedro et al., 2018, He and Wang, 2021). Unlike other metrics, such as the Pinball loss function, CRPS does not pay close attention to any specific probability distribution point but considers the distribution of the predictions. CRPS is defined in Eq. (16), which describes the difference between the observation and the prediction distributions. The smaller the value of CRPS, the better the model performance for probability forecasting:

$$PICP = \frac{1}{T} \sum_{i=1}^T Y \times 100\% \quad (14)$$

$$s.t. Y = \begin{cases} 1, & y_i \in [L_{\hat{y}_i}, U_{\hat{y}_i}] \\ 0, & y_i \notin [L_{\hat{y}_i}, U_{\hat{y}_i}] \end{cases}$$

$$PINAW = \sum_{i=1}^T \frac{U_{\hat{y}_i} - L_{\hat{y}_i}}{TD} \quad (15)$$

$$crps\left(F_i, y_i\right) = \int_{-\infty}^{+\infty} \left(F(\hat{y}_i) - H(\hat{y}_i - y_i)\right)^2 d\hat{y}_i \quad (16)$$

where  $Y$  is a Boolean variable equal to 1 when the observation value  $y_i$  is in the range of prediction value  $[L_{\hat{y}_i}, U_{\hat{y}_i}]$ , otherwise assigned to 0.  $L_{\hat{y}_i}$  and  $U_{\hat{y}_i}$  are the minimum and maximum of the prediction  $\hat{y}_i$ , respectively.  $D$  denotes the difference between the maximum and minimum in the procedure of QR. In CRPS,  $F(\hat{y})$  is the cumulative distribution function of prediction, and  $H$  denotes the Heaviside function as defined in Eq. (17).



Fig. 5. (a) The daily crude oil prices. (b) The weekly crude oil prices.

The average score of  $crps(F_i, y_i)$  is the evaluation metric of CRPS, as shown in Eq. (18).

$$H(\hat{y}_i - y_i) = \begin{cases} 1, & \hat{y}_i \geq y_i \\ 0, & \text{otherwise} \end{cases} \quad (17)$$

$$CRPS = \frac{1}{T} \sum_{i=1}^T crps(F_i, y_i) \quad (18)$$

To further verify whether the prediction performance of the proposed modelling framework is significantly better than that of the four selected models, Friedman and post hoc Nemenyi tests are performed on the crude oil price forecasting of the proposed model and other comparative models. A full description of the Friedman and post hoc Nemenyi tests is provided elsewhere (Demšar, 2006, Friedman, 1940). Briefly, the Friedman test is a non-parametric statistical test typically used to detect significant differences between the results of two or more models (Deng et al., 2023). In the Friedman test, all the tested models are ranked from best to worst according to the evaluation metrics, such as MAPE and CRPS. The rank of the  $j$ th of  $k$  models on the  $i$ th experiment of  $L$  experiment results can be devoted  $r_i^j$ . The average ranks of the  $L$  experiment results is shown in Eq. (19):

$$R_j = \frac{1}{L} \sum_{i=1}^L r_i^j \quad (19)$$

where  $L$  denotes the number of experiments for each model. Under the null hypothesis, the average rank of each model is equivalent. When  $L$  and  $k$  are sufficiently large, the Friedman statistic is distributed according to  $\chi_F^2$  approximation with degrees of freedom  $k - 1$ :

$$\chi_F^2 = \frac{12L}{k(k+1)} \left[ \sum_j R_j^2 - \frac{k(k+1)^2}{4} \right] \quad (20)$$

where  $k$  is the number of models in the comparison. Iman and Davenport

(1980) indicated that the abovementioned Friedman test is overly conservative and obtained the superior Friedman statistic shown in Eq. (21), which is distributed according to F-approximation with degrees of freedom  $k - 1$  and  $(k - 1)(L - 1)$ :

$$F_F = \frac{(L - 1)\chi_F^2}{L(k - 1) - \chi_F^2} \quad (21)$$

The critical value of  $F(k - 1, (k - 1)(L - 1))$  at the significance level of 0.05 can be obtained from statistical tables. If  $F_F$  is greater than  $F(k - 1, (k - 1)(L - 1))$ , the null hypothesis is rejected and a post hoc Nemenyi test is required to conduct further pairwise comparison (Ma et al., 2022). All models are compared based on the critical difference (CD) shown in Eq. (22):

$$CD = q_\alpha \sqrt{\frac{k(k+1)}{6L}} \quad (22)$$

where  $q_\alpha$  is determined according to the Studentised range statistic (Ma et al., 2022, Lessmann et al., 2008). If the difference in average ranks resulting from a pairwise comparison exceeds the CD, this suggests there is a significant difference.

#### 4.3. Experimental implementation

To validate and justify the probability prediction performance of the proposed model, five models were considered for comparison purposes. The SVR algorithm has a solid theoretical foundation and was chosen as the ML representation to be compared with DL models. The EEMD-CNN, EEMD-BiLSTM and CNN-BiLSTM (without decomposition), all of which are extensively used in the energy market, were chosen as three benchmark model representatives for comparison with the proposed hybrid model. The following is a detailed description of how these algorithms were implemented.

The proposed EEMD-CNN-BiLSTM-QR hybrid model is performed in

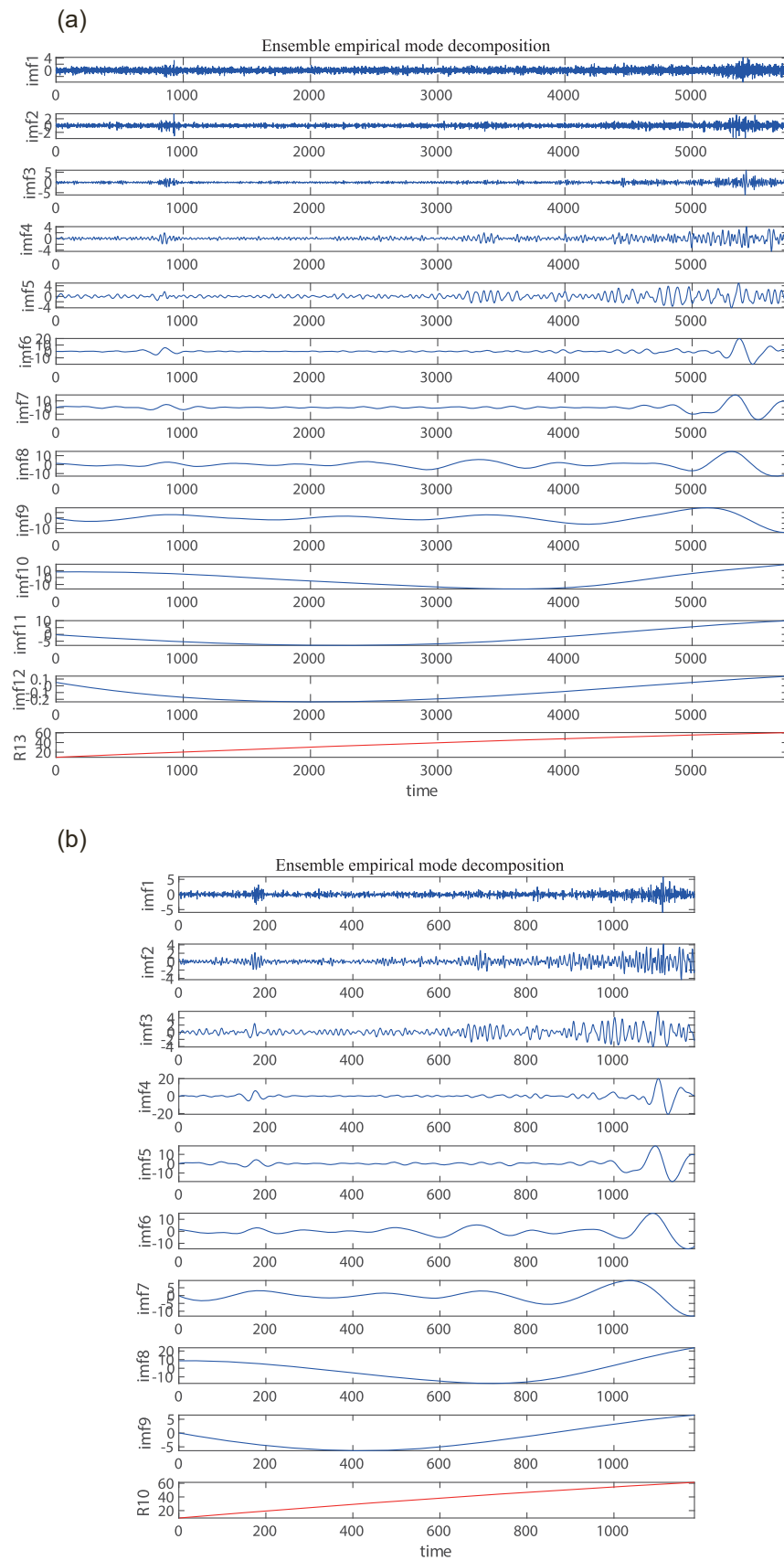


Fig. 6. (a) Daily crude oil price data decomposition (Brent). (b) Weekly crude oil price data decomposition (Brent).

**Table 4**  
RMSE and MAPE for the testing set of each sample.

| Data set | Model           | Daily  |          |               | Weekly |          |               |
|----------|-----------------|--------|----------|---------------|--------|----------|---------------|
|          |                 | RMSE   | MAPE (%) | Running time* | RMSE   | MAPE (%) | Running time* |
| Brent    | EEMD-SVR        | 4.3428 | 4.5058   | 77.92         | 2.6718 | 3.8490   | 12.92         |
|          | EEMD-BiLSTM     | 2.1119 | 3.0148   | 1734.7        | 2.1819 | 3.0735   | 286.0         |
|          | EEMD-CNN        | 2.2875 | 3.4101   | 79.61         | 2.2156 | 2.9098   | 29.55         |
|          | CNN-BiLSTM      | 2.5415 | 3.4785   | 127.02        | 3.0074 | 3.8380   | 29.66         |
|          | EEMD-CNN-BiLSTM | 1.4278 | 1.7517   | 1674.11       | 1.8621 | 2.6685   | 300.84        |
| WTI      | EEMD-SVR        | 3.6266 | 4.3705   | 74.83         | 2.5616 | 4.0145   | 13.71         |
|          | EEMD-BiLSTM     | 1.6310 | 2.0272   | 1619.23       | 1.8327 | 2.8837   | 297.10        |
|          | EEMD-CNN        | 1.7608 | 2.4030   | 81.71         | 1.8747 | 2.9215   | 29.68         |
|          | CNN-BiLSTM      | 2.4194 | 2.8534   | 131.22        | 2.6759 | 4.3720   | 30.81         |
|          | EEMD-CNN-BiLSTM | 1.3954 | 1.8661   | 1729.17       | 1.4390 | 2.1581   | 309.86        |

Note: Bold text indicates the lowest value of each error measure for each dataset.  
\*Running time was recorded seconds.

the environment of MATLAB R2021b platform based on the following graphics processing unit (GPU): (CPU: Intel(R) Core (TM) i7-8700 K @3.7 GHZ; RAM: 16 GB; GPU: NVIDIA GeForce GTX 1070), where QR plays a crucial role in PDF. First, data were pre-processed using a decomposition approach using the EEMD technique. This technique is used to capture the intrinsic features of original signals in an adaptable manner according to time and frequency amplitude. In the function of EEMD, 'Nstd' indicates the ratio of the standard deviation of the added Gaussian white noise to that of the original time series, which is set to 0.2. NE is the ensemble number for the EEMD, which adds Gaussian white noise to the original data and is set to 100. The components obtained are split into training and testing sets, which are input into the network structure. Second, the procedure for modelling includes two modules: CNN and BiLSTM. The 1D-CNN is used to extract spatial information about crude oil prices. The time step of the input variable is set to 5, meaning that the records of five historical data points are used to forecast the crude oil price at the sixth time step. Meanwhile, we create a layer with 16 convolution kernels of size  $5 \times 5$ , a layer with 32 convolution kernels of size  $5 \times 5$  and two 1-D max pooling layers with a pool size of 3, a stride of 1 and padding of size = 1 with padding-mode = 'same' on both the left and right of the input. In addition, the BiLSTM is employed to capture the non-linear temporal long-term inherent relationship at the given time step. The BiLSTM module includes two layers: the first has 200 hidden units and the second has 150 hidden units. To reduce the likelihood of over-fitting, two dropout layers are added alternately at the ends of two BiLSTM layers. The dropout rate is set to 0.2; that is, 20 % of the neurons are dropped out. During the training process, adaptive moment estimation (Adam) is used as the optimiser (Hou et al., 2022). The max epochs and mini-batch sizes are 150 and 16, respectively.

The SVR method based on EEMD is also implemented in the environment of MATLAB R2021b. However, the three hyper-parameters (C,  $\gamma$ ,  $\epsilon$ ) in SVR are set to 64, 2 and 0.001, respectively, based on the work proposed by Huang et al. (2022). Moreover, another three benchmark models are performed in the same environment as the above. The network layer and training options of the network structure in each benchmark model are set in the same way as those of the proposed model. To guarantee the reliability of comparisons, the average performance of each model is used for comparison in the testing set.

## 5. Experimental results

### 5.1. Decomposition analysis

Daily and weekly crude oil prices were taken as daily and weekly datasets, respectively, from Brent and WTI. As can be seen in Fig. 5(a) and Fig. 5(b), each sample has transitioned from a pattern that was reasonably steady in the past to one that presently fluctuates dramatically and significantly in the complicated energy market. It is important

to note that the fluctuations in crude oil prices provide a highly stochastic situation, particularly in recent decades. As discussed above, the most important factors driving crude oil prices are oil consumption, production, inventory, finances, politics and significant global unrest (Gupta and Nigam, 2020, Lin et al., 2020). To improve the accuracy of probability forecasting by capturing the inherent features of crude oil prices, this study suggest that data preprocessing be included in the construction of the model (Wu et al., 2019).

In addition, EEMD is a noise-assisted data analysis approach that extracts  $N$  *imfs* and an adaptively calculated residual component from the original signals. The value of  $N$  corresponds to an integer equal to  $\log_2(T) - 1$  (Wu and Huang, 2009). That is, the number of components obtained is unequal if the length of the original signals has significant variation. In the model estimation stage for the proposed models, all the samples from training sets are decomposed at one time and used for model estimation. Finally, the achieved model based on training sets is tested on the hold-out sample as the decomposition is repeated with subsequent data added.

Because of space constraints, only the findings from the decomposition for the Brent training sets are shown here. Fig. 6(a) and Fig. 6(b) illustrate how daily crude oil prices are decomposed into 12 *imfs* and a residue and how weekly crude oil prices are decomposed into 9 *imfs* and a residue. It is not difficult to demonstrate that with the exception of the residue, all components vibrate up and down at zero. Nevertheless, the spectrum of  $N$  *imfs* is asymmetric, which has specific relevance from a physical standpoint. In addition, the residue  $Rn$  is a monotonically growing positive number, which indicates the crude oil price is now exhibiting an upward trend because of the limited nature of the resources that do not restore themselves.  $Rn$  denotes the primary trend, which is unaffected by the factors influencing the energy market. Despite this, forecasting the residue component is crucial for the study's short-term forecasting. The low-frequency components retrieved from the *imfs* indicate the crude oil prices are influenced by certain stochastic factors to have a large short-term effect. For the sake of decision making by politicians or financial managers, having an accurate prediction of the components is particularly crucial. Moreover, the high-frequency components have mini-amplitude compared to low-frequency components. Thus, accurate forecasting of high-frequency components is crucial for the decision-making process in industrial manufacturing.

### 5.2. Statistical analysis

In this section, the experimental results for both determined and probability forecasting are presented. One set of results is used to validate the accuracy of the prediction made by the EEMD-CNN-BiLSTM-QR modelling framework, while the other set is used to quantify the non-determinacy of the prediction process. The statistical metrics for both determined and probability forecasting are computed. Each model is run 20 times, resulting in 20 sets of MAPE, RMSE, Pinball, PICP, PINAW and

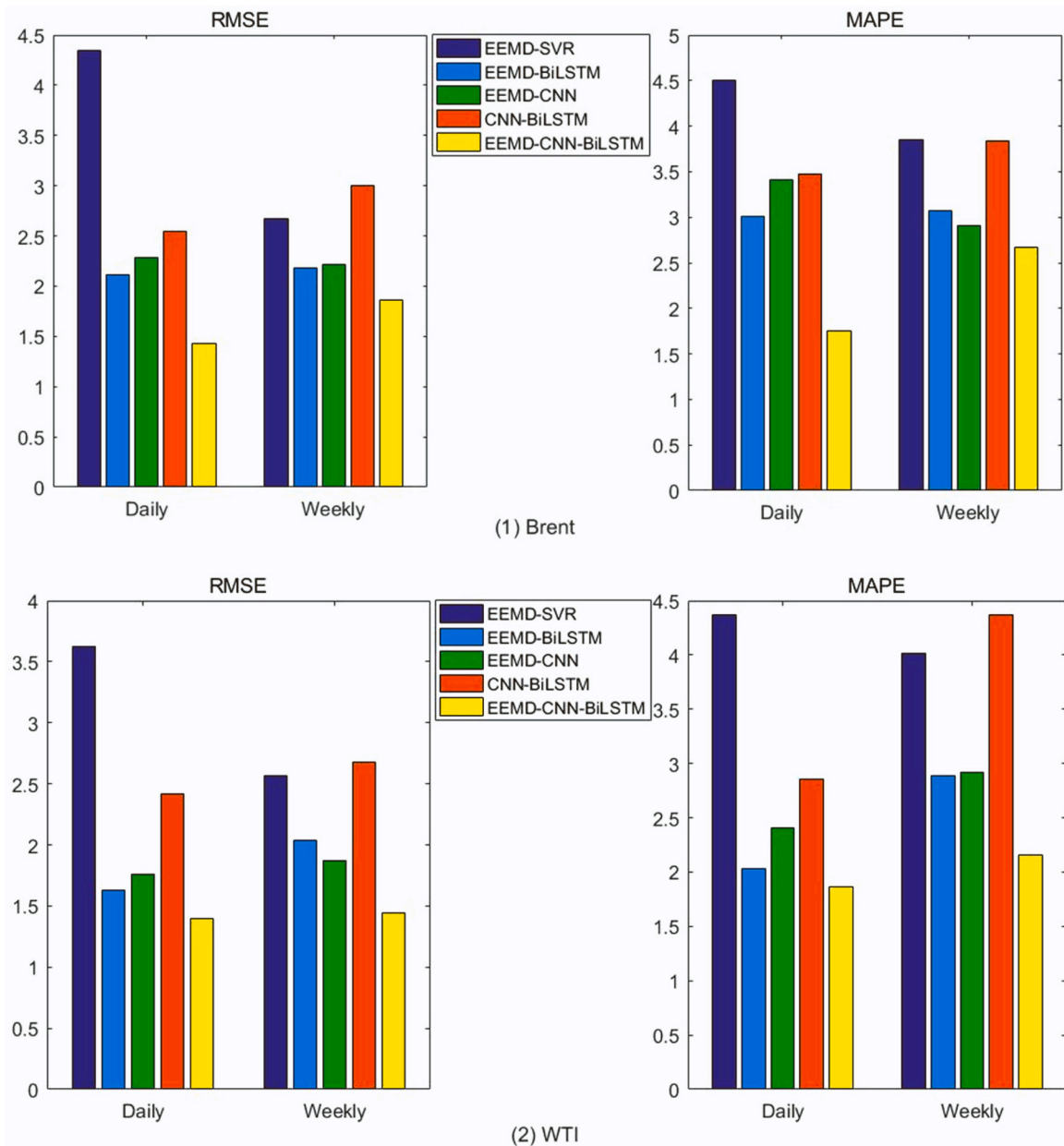


Fig. 7. Comparison of RMSE and MAPE for the five models.

CRPS values. The mean of the experimental outcomes is computed to facilitate comparison and analysis.

#### 5.2.1. Determined forecasting statistical analysis

The evaluation metrics for determined forecasting are used here to check whether the proposed model generates accurate predictions. In Table 4, the RMSE and MAPE of five different models for each sample are presented. It is clear that the proposed EEMD-CNN-BiLSTM model has the highest degree of accuracy in its predictions among all models. For further quantitative analysis, the amount of time spent running each model is recorded in Table 4.

EEMD is used to extract the intrinsic characteristics of the complex crude oil price time series to improve prediction performance. Table 4 validates the effectiveness of EEMD time-frequency decomposition in comparison with the proposed EEMD-CNN-BiLSTM model and the CNN-BiLSTM model (without the decomposition of the EEMD technique). Taking the daily dataset in the Brent market as an example, the RMSE and MAPE of EEMD-CNN-BiLSTM are 1.4278 and 1.7517, which are

reduced by 43.82 % and 49.64 % compared with CNN-BiLSTM, respectively. The minimum error of the proposed model was also obtained, and the RMSE and MAPE of EEMD-CNN-BiLSTM are reduced by 38.08 %, 42.32 %, 46.19 % and 30.47 %, 34.60 %, 50.63 %, respectively, when compared to CNN-BiLSTM.

It is worth noting that the performance of EEMD-CNN-BiLSTM is superior to that of the other three tested and compared models evaluated in both the Brent and WTI markets. Considering the weekly dataset from the WTI market as an example, the RMSE and MAPE of EEMD-CNN-BiLSTM are 1.4390 and 2.1581, which is reduced by 43.82 %, 21.48 % and 23.23 % (in the context of RMSE) and 46.24 % and 26.13 % (in the context of MAPE) when compared to EEMD-SVR, EEMD-BiLSTM and EEMD-CNN, respectively. To show the results more intuitively, histograms for RMSE and MAPE for the daily and weekly datasets of the Brent and WTI market are plotted in Fig. 7. Among the models using the decomposition technique of EEMD, the experimental result for the EEMD-SVR is the worst in terms of RMSE and MAPE, which verifies that DL models are superior to ML models for



**Table 5**

Friedman and post hoc Nemenyi test results on MAPEs.

| Criteria       | Models          | Brent   |         | WTI     |         |
|----------------|-----------------|---------|---------|---------|---------|
|                |                 | daily   | weekly  | daily   | weekly  |
| Average MAPE   | EEMD-SVR        | 4.5058  | 3.8490  | 4.3705  | 4.0145  |
|                | EEMD-BiLSTM     | 3.0148  | 3.0735  | 2.0272  | 2.8837  |
|                | EEMD-CNN        | 3.4101  | 2.9098  | 2.4030  | 2.9215  |
|                | CNN-BiLSTM      | 3.4785  | 3.8380  | 2.8534  | 4.3720  |
|                | EEMD-CNN-BiLSTM | 1.7517  | 2.6685  | 1.8661  | 2.1581  |
| Average rank   | EEMD-SVR        | 4.4500  | 3.9500  | 4.1650  | 3.9500  |
|                | EEMD-BiLSTM     | 2.6000  | 3.1650  | 2.8500  | 2.7500  |
|                | EEMD-CNN        | 3.2500  | 2.9500  | 3.1000  | 2.7000  |
|                | CNN-BiLSTM      | 3.4750  | 3.4150  | 3.7500  | 4.3650  |
|                | EEMD-CNN-BiLSTM | 1.2250  | 1.6150  | 1.2500  | 1.3250  |
| $\chi^2_F$     |                 | 45.6100 | 28.7414 | 45.6378 | 50.1108 |
| $F_F$          |                 | 25.1989 | 10.6536 | 25.2347 | 31.8545 |
| Critical value |                 | 2.4920  | 2.4920  | 2.4920  | 2.4920  |
| $q_{0.05}$     |                 | 2.728   | 2.728   | 2.728   | 2.728   |
| CD             |                 | 1.364   | 1.364   | 1.364   | 1.364   |

crude oil price forecasting. Furthermore, among all the models of DL using the decomposition of EEMD, the hybrid EEMD-CNN-BiLSTM model is superior to the benchmark models, which demonstrates the superiority of the proposed model integrating the feature extraction of CNN with the long-term dependence learning of BiLSTM. Moreover, the running time of each model explains the accumulation of hidden layers of DL. They need more time to learn and train the model.

To further examine whether the prediction performance of the proposed EEMD-CNN-BiLSTM model is significantly better than that of the other models compared, Friedman and post hoc Nemenyi tests are conducted on the crude oil price-determined forecasting of the proposed algorithm and other comparative methods. The Friedman test performs the 20 sets of MAPE experimental data from five different models and post hoc Nemenyi tests, which can comprehensively evaluate the proposed model in crude oil price time series. As shown in Table 5, the Friedman and post hoc Nemenyi test results show that the proposed EEMD-CNN-BiLSTM model is significantly different ( $p < 0.05$ ) from the other four comparative methods in terms of determined forecasting performance.

With five methods and 20 sets of experimental data,  $F_F$  is distributed according to F-approximation with degrees of freedom  $5 - 1 = 4(5 - 1)(20 - 1) = 76$ . From statistical tables, the critical value  $F(4, 76)$  at a significance level of 0.05 is 2.4920; thus the null hypothesis is rejected. The Nemenyi test is used to conduct further analysis for pairwise comparisons. According to the Studentised range statistic,  $q_{0.05}$  is 2.728, and the corresponding CD is 1.364. In Table 6, the average ranks of the five models are computed. The corresponding CD diagrams represent the performance differences between algorithms according to the average ranks of the five models for each dataset.

**Table 6**

Pinball, CRPS, PINAW and PICP for the testing set for each sample.

| Dataset | Model              | Daily   |        |        |          |                    | Weekly  |        |        |          |                    |
|---------|--------------------|---------|--------|--------|----------|--------------------|---------|--------|--------|----------|--------------------|
|         |                    | Pinball | CRPS   | PINAW  | PICP (%) | Running time (GPU) | Pinball | CRPS   | PINAW  | PICP (%) | Running time (GPU) |
| Brent   | EEMD-SVR-QR        | 1.6724  | 2.5823 | 0.3214 | 85.4285  | 482.1526           | 1.8295  | 2.9230 | 0.3941 | 82.3126  | 253.9523           |
|         | EEMD-BiLSTM-QR     | 1.2056  | 2.0516 | 0.1806 | 100.000  | 16815.2640         | 1.4892  | 2.5368 | 0.2253 | 100.000  | 2918.9553          |
|         | EEMD-CNN-QR        | 1.1729  | 1.9811 | 0.1775 | 100.000  | 554.1363           | 1.2641  | 2.1670 | 0.2033 | 99.1852  | 234.1422           |
|         | CNN-BiLSTM-QR      | 1.2994  | 2.1878 | 0.1621 | 78.1055  | 1602.4191          | 1.4184  | 2.2778 | 0.1782 | 76.6892  | 356.4237           |
|         | EEMD-CNN-BiLSTM-QR | 1.0308  | 1.7602 | 0.1330 | 100.000  | 14461.6212         | 1.2370  | 2.0759 | 0.1869 | 100.000  | 2642.1644          |
| WTI     | EEMD-SVR-QR        | 1.5926  | 2.6518 | 0.2891 | 82.8236  | 475.9252           | 1.8043  | 2.7845 | 0.3052 | 86.1983  | 237.9126           |
|         | EEMD-BiLSTM-QR     | 1.1870  | 2.1497 | 0.2253 | 99.9665  | 16747.7242         | 1.2387  | 2.0836 | 0.2154 | 100.000  | 2980.3693          |
|         | EEMD-CNN-QR        | 1.1918  | 1.8323 | 0.1551 | 99.9665  | 549.3105           | 1.2590  | 2.1601 | 0.1880 | 99.3752  | 235.2111           |
|         | CNN-BiLSTM-QR      | 1.2386  | 2.1761 | 0.1221 | 73.4851  | 1682.9660          | 1.3234  | 2.2423 | 0.1732 | 73.6487  | 360.0350           |
|         | EEMD-CNN-BiLSTM-QR | 1.0211  | 1.7062 | 0.1256 | 99.9665  | 15920.8411         | 1.1152  | 1.8879 | 0.1811 | 100.000  | 2787.4212          |

Note: Bold text indicates the highest value of each error measure for each dataset

Fig. 8 shows the result of pairwise comparisons of the five methods using Nemenyi's post hoc test. Each horizontal line segment depicts the magnitude of the mean rank, with its width reflecting the magnitude of the CD value. The black vertical line is used to determine whether the difference in mean rank resulting from a pairwise comparison exceeds the CD, which suggests a significant difference. If the vertical line does not cross two horizontal lines simultaneously, this indicates a significant difference between the two methods. As shown in Fig. 8, EEMD-CNN-BiLSTM is superior to other tested models' ability to forecast crude oil prices. The performance of EEMD-CNN-BiLSTM is significantly better than that of EEMD-SVR and CNN-BiLSTM, indicating that the hybrid modelling framework with time-frequency techniques and DL NNs can effectively provide a comprehensive evaluation of future trends in crude oil prices. Meanwhile, EEMD-CNN-BiLSTM performs significantly better than EEMD-CNN, and EEMD-BiLSTM verified that the temporal-spatial features extracted could improve crude oil price prediction performance.

### 5.2.2. Undetermined forecasting statistical analysis

To quantify the non-determinacy of prediction, the evaluation metrics of undetermined forecasting are used to validate the accuracy of probability density forecasts in this paper. The Pinball, CRPS, PINAW and PICP for the five models for each sample are presented in Table 6. The prediction results for 21 quantiles ranging from 0.01 to 0.99 are used to construct the PIs for the probability density forecast of crude oil prices.

Table 6 compares all models, including Pinball, CRPS, PINAW and PICP for various datasets. Without the time-frequency decomposition of EEMD, the CNN-BiLSTM-QR model has the lowest PICP for each sample among all the models. This strategy further proves that EEMD is suitable for the probability density forecast of crude oil prices. Although the CNN-BiLSTM-QR has the smallest PINAW, the reliability of its PIs is rather worse than that of other models. Moreover, compared with the EEMD-SVR-QR model, the PICP values for the three DL models based on EEMD exceed 99 % for each dataset. This technique indicates that the DL models can achieve higher reliability because of the tremendous self-adaptive learning ability of DL. Taking the daily crude oil prices in the Brent market as an example, the reliability of each model based on the DL technique is 17.06 % higher than that of the EEMD-SVR-QR. The three models based on DL have very similar PICP and different PINAW. Therefore, the proposed model obtains a better CRPS than the other two DL models, which indicates the superiority of integration of the convolution and pooling operation of CNN with the bidirectional memory module of BiLSTM. The experimental results demonstrate that CRPS can be used to assess the resolution of performance for probability density forecast, comprehensively considering both sharpness and reliability. Taking the daily crude oil prices in the WTI market as an example, the EEMD-CNN-BiLSTM-QR model has the lowest CRPS—35.66 %, 20.63 %, 6.89 % and 21.59 % lower than that for EEMD-SVR-QR, EEMD-BiLSTM, EEMD-CNN-QR and CNN-BiLSTM-QR,

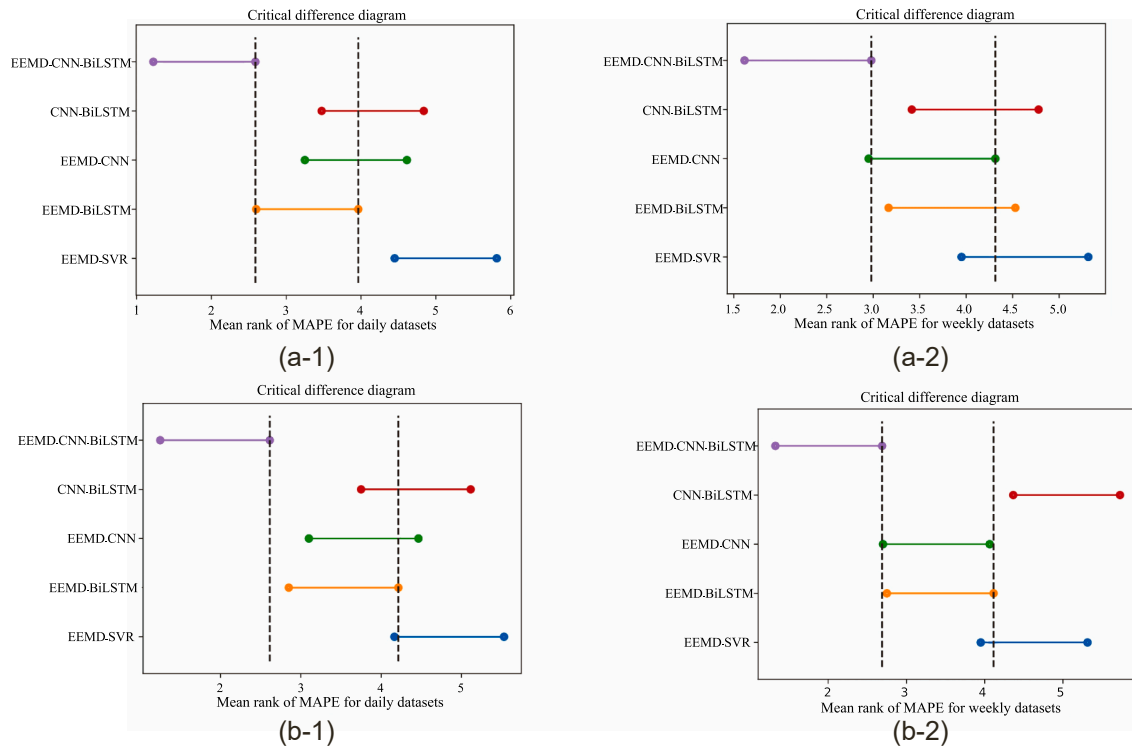


Fig. 8. Pairwise comparisons of MAPE using post hoc Nemenyi test.

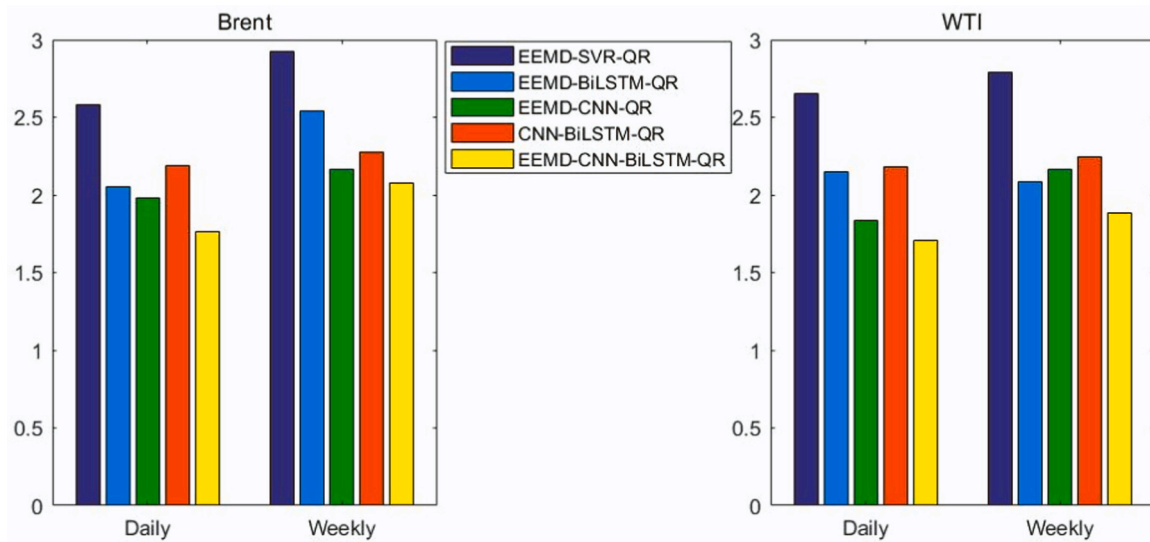


Fig. 9. Comparison of CRPS for the five models.

respectively. As shown in Fig. 9, the proposed model achieves the best CRPS of probability density forecast for each dataset among the selected models. Furthermore, the Pinball loss is an important metric to evaluate the accuracy of the quantile forecast. As shown in Table 6, taking the weekly crude oil prices in the WTI market as an example, the EEMD-CNN-BiLSTM-QR model has the smallest Pinball loss—lower by 38.19 %, 9.97 %, 11.42 % and 15.73 %, respectively, than the other four models. This phenomenon also occurs in other datasets that explain how the hybrid EEMD-CNN-BiLSTM-QR model can effectively capture the long-term dependence relationship of the time series under different quantiles to improve the undetermined forecasting of crude oil prices.

Although the running time for the proposed model is quite long, it is not the longest. The accumulation of hidden layers of DL means that

more time is needed to learn and train the probability prediction model. Therefore, the time for implementation of the four models using the DL technique was longer than that for EEMD-SVR-QR. Moreover, the components decomposed by EEMD are needed to model and forecast, respectively, so the CNN-BiLSTM-QR takes less time than do EEMD-BiLSTM-QR, EEMD-CNN-QR and the model proposed in this study. Compared to models based on the BiLSTM technique, the EEMD-CNN-QR model requires less time in implementation, which indicates the superiority of the convolution and pooling operations in the CNN for extracting features of crude oil prices. Fortunately, the EEMD-CNN-BiLSTM-QR is implemented in less time than the EEMD-BiLSTM-QR. This is because utilising the CNN can effectively reduce the running time of the proposed hybrid EEMD-CNN-BiLSTM-QR model.

**Table 7**  
Friedman and post hoc Nemenyi test results on CRPSs.

| Criteria       | Model              | Brent    |         | WTI      |         |
|----------------|--------------------|----------|---------|----------|---------|
|                |                    | Daily    | Weekly  | Daily    | Weekly  |
| Average CRPSs  | EEMD-SVR-QR        | 2.5823   | 2.6518  | 2.923    | 2.7845  |
|                | EEMD-BiLSTM-QR     | 2.0516   | 2.1497  | 2.5368   | 2.0836  |
|                | EEMD-CNN-QR        | 1.9811   | 1.8323  | 2.167    | 2.1601  |
|                | CNN-BiLSTM-QR      | 2.1878   | 2.1761  | 2.2778   | 2.2423  |
|                | EEMD-CNN-BiLSTM-QR | 1.7602   | 1.7062  | 2.0759   | 1.8879  |
| Average rank   | EEMD-SVR-QR        | 4.3750   | 3.9520  | 4.355    | 3.7875  |
|                | EEMD-BiLSTM-QR     | 3.0950   | 2.9800  | 3.239    | 3.1500  |
|                | EEMD-CNN-QR        | 2.9830   | 2.7900  | 2.962    | 2.8500  |
|                | CNN-BiLSTM-QR      | 3.8150   | 3.745   | 3.657    | 3.6500  |
|                | EEMD-CNN-BiLSTM-QR | 1.550    | 1.286   | 1.6135   | 1.397   |
| $\chi^2_F$     |                    | 76.5973  | 23.6930 | 73.6610  | 21.3141 |
| $F_F$          |                    | 427.7057 | 7.9949  | 220.7843 | 6.9006  |
| Critical value |                    | 2.4920   | 2.4920  | 2.4920   | 2.4920  |
| $q_{0.05}$     |                    | 2.728    | 2.728   | 2.728    | 2.728   |
| CD             |                    | 1.364    | 1.364   | 1.364    | 1.364   |

To further examine whether the prediction performance of the proposed hybrid modelling framework is significantly better than that of the other probability forecasting models compared, Friedman and post hoc Nemenyi tests are performed on the crude oil price undetermined forecasting of the proposed algorithm and other comparative methods. To provide a comprehensive evaluation, 20 sets of CRPS experimental results from five different models are analysed using the Friedman and post hoc Nemenyi tests. Table 7 shows that the proposed EEMD-CNN-BiLSTM-QR model performs significantly better ( $p < 0.05$ ) than the

other four comparative models in terms of the undetermined prediction evaluation metric CRPS.

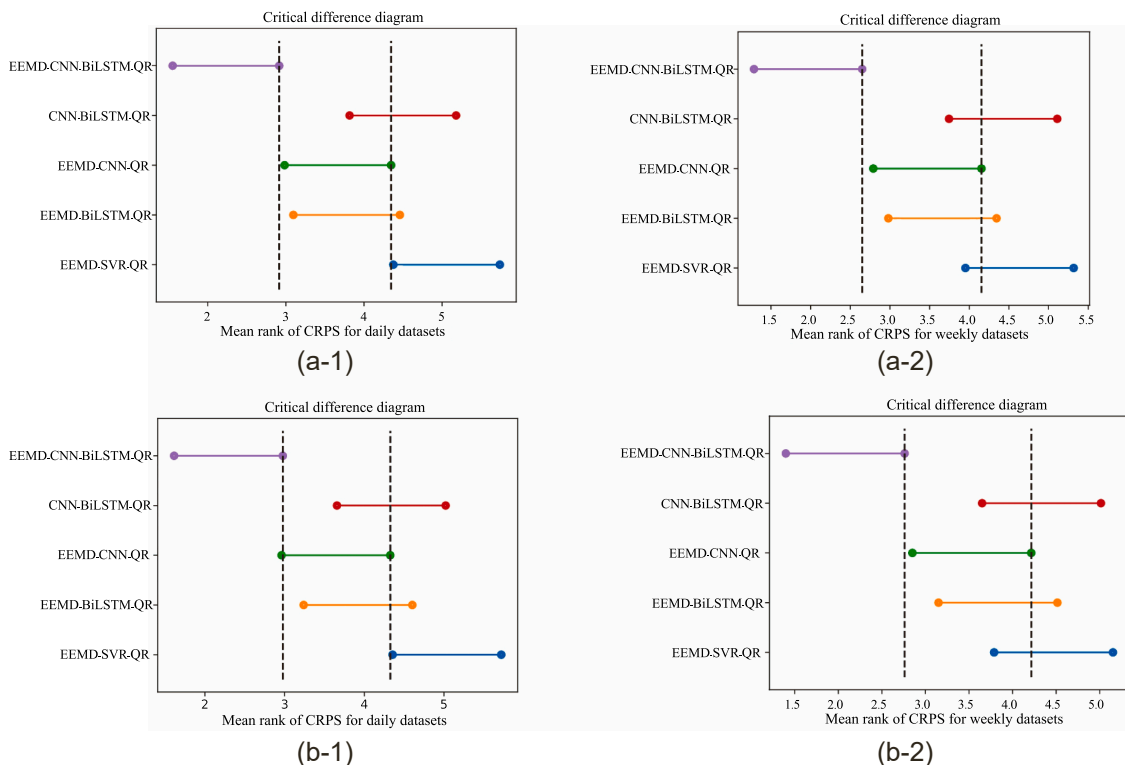
As shown in Fig. 10, four CD diagrams present the results of pairwise comparisons of five methods using the post hoc Nemenyi test. The first black vertical line does not cross two horizontal lines simultaneously, so the proposed EEMD-CNN-BiLSTM-QR outperforms the other comparative probability forecasting models. The performance of EEMD-CNN-BiLSTM-QR is significantly better than that of CNN-BiLSTM-QR, indicating that EEMD can effectively extract the complex non-linearity features of crude oil prices. Similarly, EEMD-CNN-BiLSTM-QR performs significantly better than EEMD-CNN-QR. EEMD-BiLSTM-QR indicates that the temporal-spatial features extracted can improve the prediction probability forecasting performance of crude oil prices. Furthermore, the mean rank based on DL is better than that based on ML, though the mean rank of CNN-BiLSTM-QR is slightly better than that for EEMD-SVR-QR.

## 6. Discussion

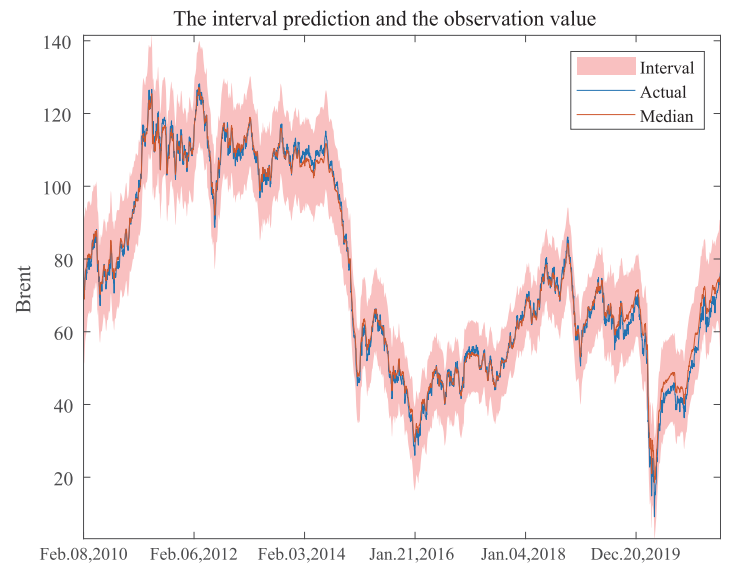
The section presents PI results and probability density curves to provide valuable information for the PDF of crude oil prices.

### 6.1. Discussion based on daily data

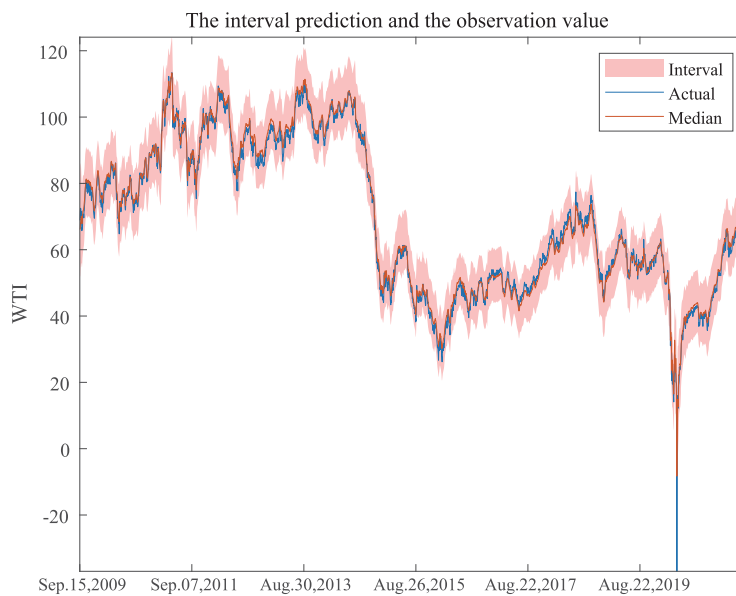
The results distribution of prediction is shown for daily crude oil prices from the Brent and WTI markets using the hybrid EEMD-CNN-BiLSTM-QR model in Fig. 11, which includes the interval prediction from 0.01 to 0.99 quantiles, the actual values and the prediction in the median. The values of PICP in Table 6 and Fig. 11 demonstrate that almost all of the actual values are in the range of PIs. The prediction in the median is close to the actual value. However, the value of PICP does not reach 100 % in WTI crude oil price forecasting, which indicates that the crude oil price on 20 April 2020 is significantly affected by uncertain factors during this period, though the prediction interval is wider at this point than at other times.



**Fig. 10.** Pairwise comparisons of CRPS using post hoc Nemenyi test.



(a)

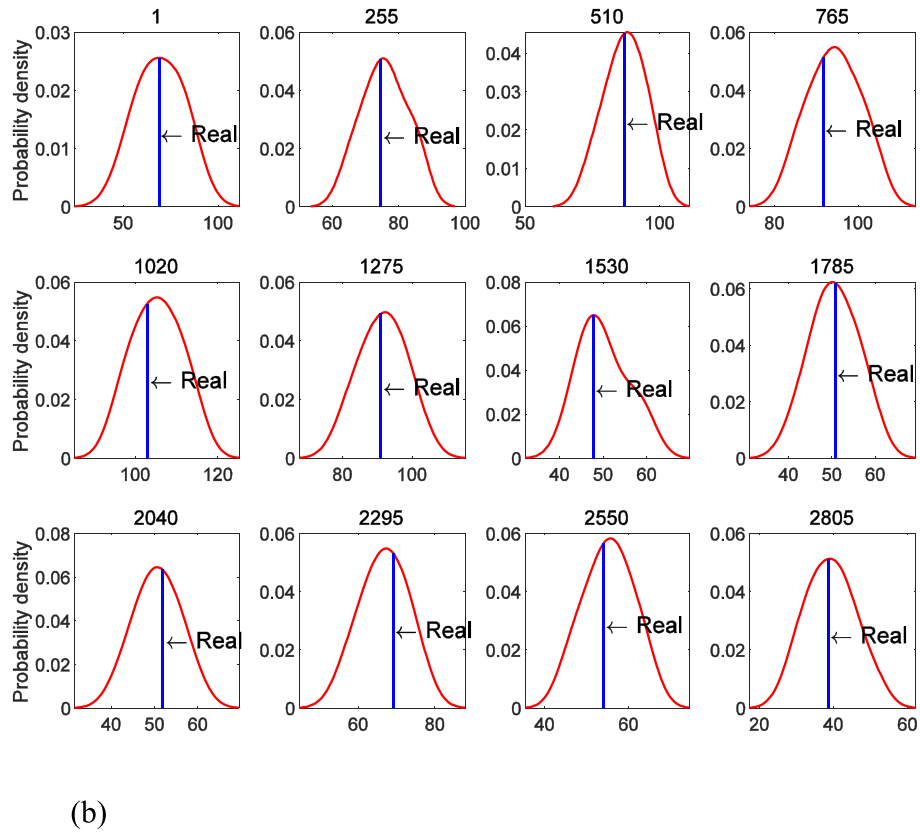
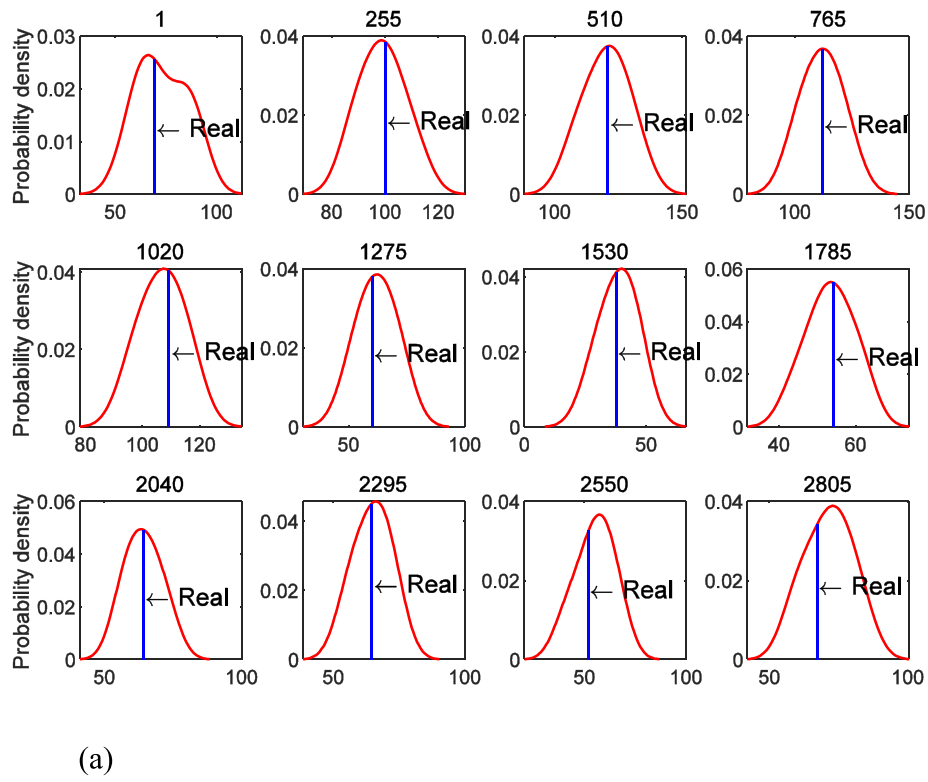


(b)

**Fig. 11.** The PI for daily crude oil prices using the proposed model.

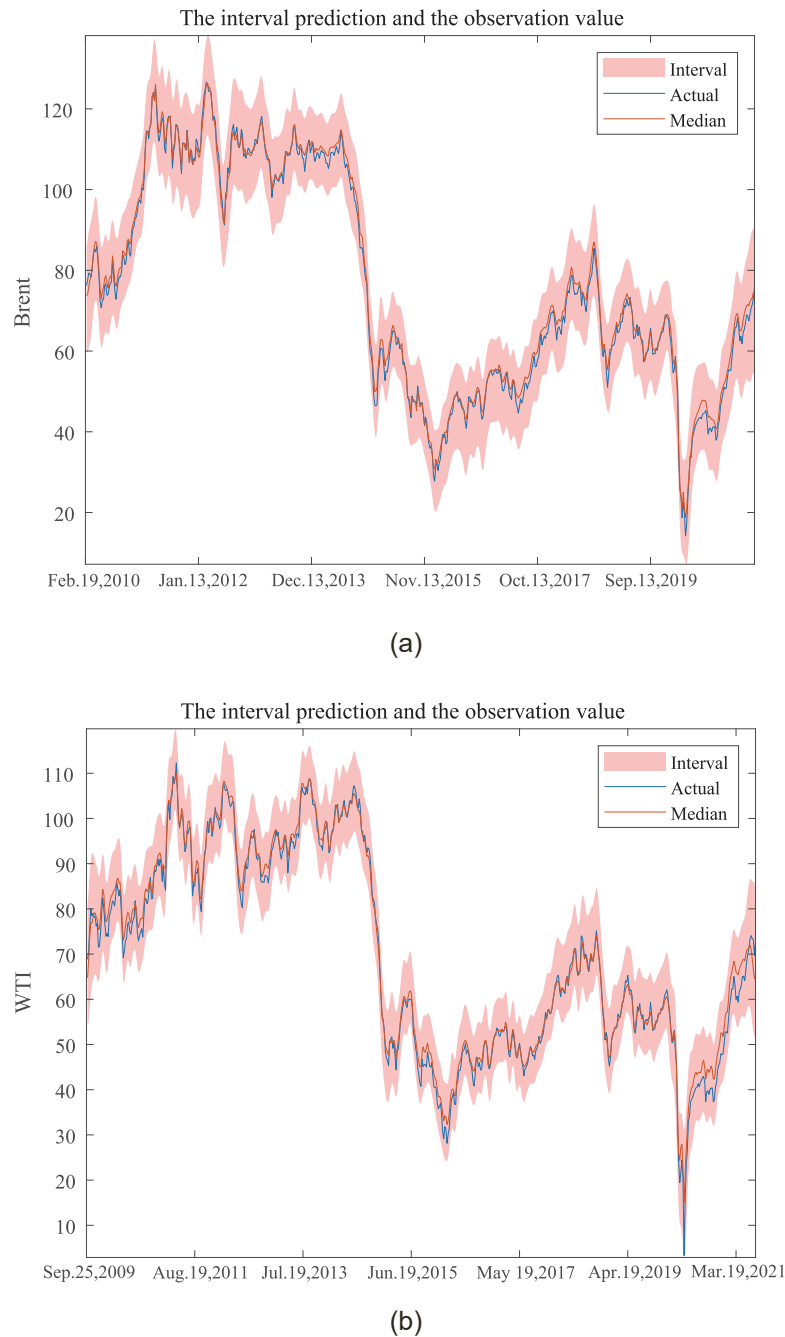
As shown in Table 6, the value of PICP using the time-frequency decomposition of EEMD is higher than that without the time-frequency decomposition of EEMD. Meanwhile, the value of PINAW using the DL NN is smaller than that with ML. The experimental results validate that the prediction intervals based on the EEMD-CNN-BiLSTM-QR model achieve better performance than the other four tested models. Furthermore, in comparison with ICEEMDAN-FE-FIG-ARIMA-IESN proposed by Zhang and Liu (2024), the value of PICP using the EEMD-CNN-BiLSTM-QR model is 9.6765 higher for daily WTI crude oil prices and 1.46 higher for daily Brent Crude Oil prices. Conversely, the value of PINAW for the EEMD-CNN-BiLSTM-QR model is 0.0111 lower for daily WTI crude oil prices and 0.3499 lower for daily Brent Crude Oil prices. These results validate the shortcoming of considering only the

temporal features and ignoring the spatial features of crude oil prices. From the prediction accuracy of the model, as shown in Table 4, the EEMD-CNN-BiLSTM model achieves better prediction results with a value of RMSE (1.4278) and MAPE (1.7517 %) in the context of Brent. For the WTI market the RMSE is 1.3954 and the MAPE is 1.8661 %. From the histograms of CRPS in Fig. 9, the proposed EEMD-CNN-BiLSTM-QR model achieves the lowest CRPS, which indicates it has the highest accuracy of probability forecasting among the five models. Meanwhile, as shown in Fig. 8 and Fig. 10, Friedman and post hoc Nemenyi tests further verify that the prediction performance of the proposed modelling framework is significantly better than that of the other four models. These results indicate that the performance of predictions based on the EEMD-CNN-BiLSTM-QR model is superior to that



**Fig. 12.** (a) Probability density distribution of 12 random days from the Brent market. (b) Probability density distribution of 12 random days from the WTI market. Probability density curves of the EEMD-CNN-BiLSTM-QR for daily crude oil prices.





**Fig. 13.** The PI for weekly crude oil prices using the proposed model.

of other tested models under different quantiles.

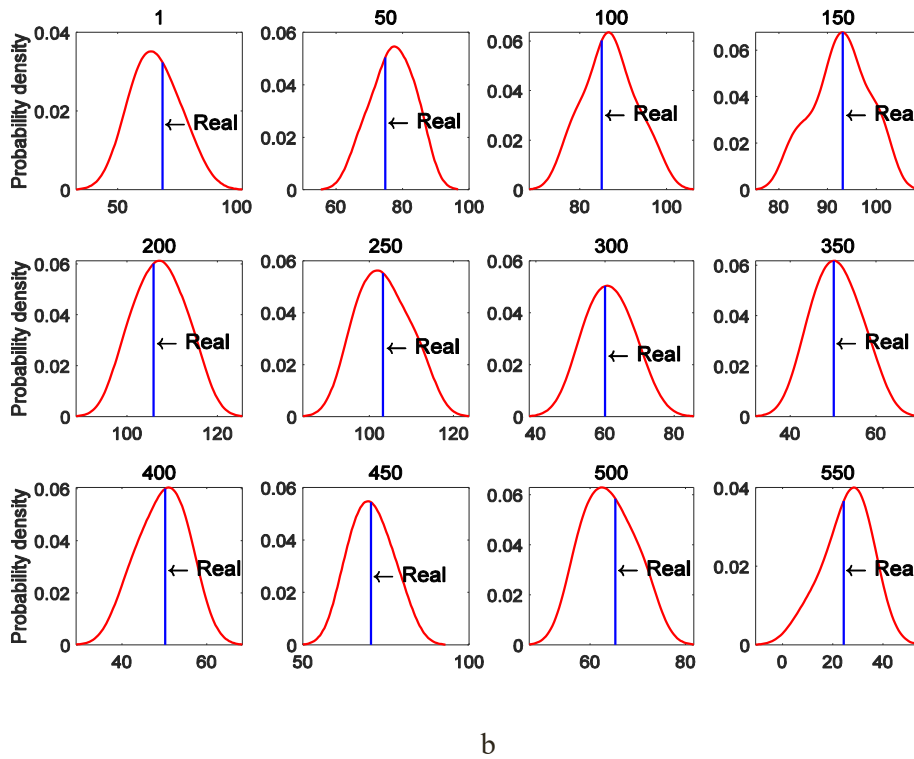
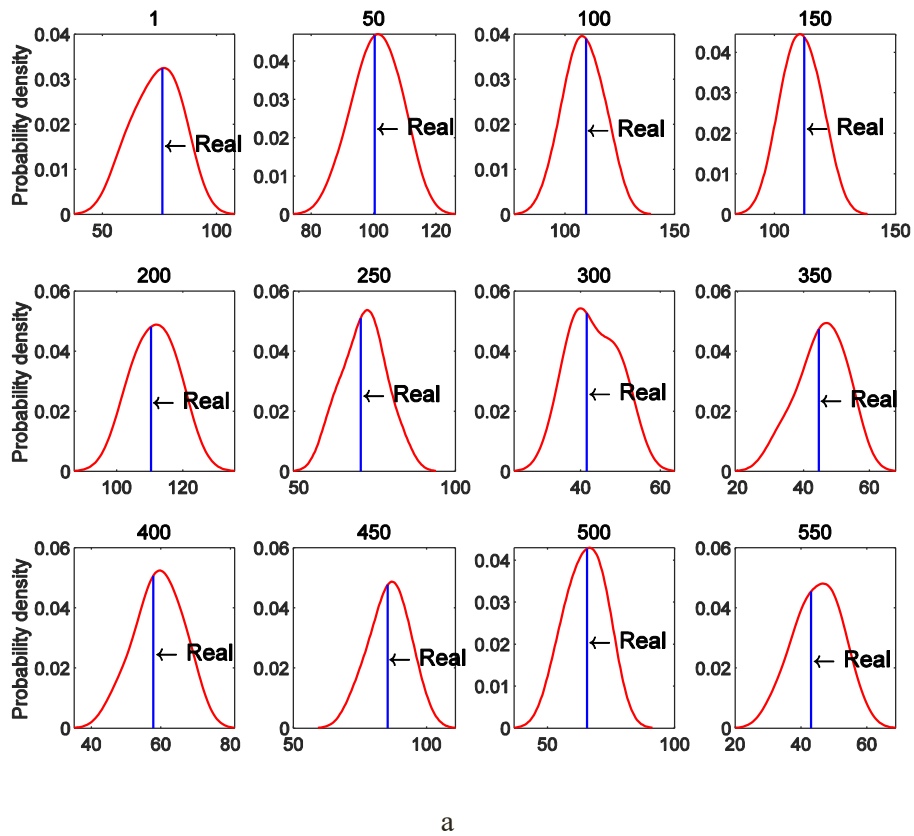
All the predictions under different quantiles are fed into KDE to establish probability density curves. The probability density curves include three aspects: the distribution of an undetermined prediction, the probability of each distribution and the probability density of each time. Each piece of information is significant for the participants in the energy market.

Fig. 12 describes the probability density curves obtained by the EEMD-CNN-BiLSTM-QR model in the 1st, 255th, 510th, 765th, 1020th, 1275th, 1530th, 1785th, 2040th, 2295th, 2550th and 2805th days. It is not difficult to find the real value of crude oil prices, which is almost within the range of the probability density distribution at the corresponding time point. The real values approximate the prediction of

maximum probability. The experimental results indicate that the probability density distribution based on the proposed model can accurately quantify the volatility of crude oil prices in undetermined forecasting.

## 6.2. Weekly data discussion

The results distribution of prediction is shown for weekly crude oil prices for the Brent and WTI markets using the proposed model in Fig. 13, which includes the interval prediction from 0.01 to 0.99 quantiles, the actual values and the prediction in the median. The value of PICP in Table 6 and Fig. 13 shows that almost all the actual values are in the range of PIs. The prediction in the median is close to the actual value. Furthermore, in comparison with the model proposed by Zhang



**Fig. 14.** (a) Probability density distribution of 12 random weeks from the Brent market. (b) Probability density distribution of 12 random weeks from the WTI market. Probability density curves of the EEMD-CNN-BiLSTM-QR for weekly crude oil prices.

and Liu (2024), the EEMD-CNN-BiLSTM-QR model shows PICP values 4.8800 higher for weekly WTI crude oil prices and 7.3200 higher for weekly Brent crude oil prices. Simultaneously, the value of PINAW for the EEMD-CNN-BiLSTM-QR model is 0.1600 lower for weekly WTI crude oil prices and 0.1369 lower for daily Brent crude oil prices. These improvements validate the performance of the proposed hybrid model.

Fig. 14 describes the probability density curves obtained by the EEMD-CNN-BiLSTM-QR model in the 1st, 50th, 100th, 150th, 200th, 250th, 300th, 350th, 400th, 450th, 500th and 550th weeks. It is not difficult to find the real value of crude oil prices, which is almost in the range of the probability density distribution at the corresponding time point. The real values approximate the prediction of maximum probability. The experimental results indicate that the probability density distribution based on the proposed model can accurately quantify the volatility of crude oil prices in undetermined forecasting.

### 6.3. Management applications

This section examines the efficacy and effectiveness of the proposed EEMD-CNN-BiLSTM-QR framework in forecasting crude oil prices. The experiment confirms the framework's capacity and versatility in predicting crude oil prices and demonstrating its practical use. Probability density forecasting can effectively capture the fluctuating degrees of volatility and non-Gaussian distribution patterns in time series data. This proposed strategy enables industry leaders to monitor trading activity crude oil prices and adaptations in fundamental traits. Moreover, probability density forecasting offers a spectrum of potential outcomes alongside their associated likelihoods, rather than simply a point estimation. Energy industries can comprehend the degree of uncertainty inherent to the forecasting. Managers may make better judgments and create risk management plans by measuring uncertainty to reduce possible negative results. Moreover, managers might assess a full spectrum of potential outcomes rather than relying solely on point forecasting, which may fail to account for unexpected risks or opportunities.

In addition, probability density forecasts provide a clear representation of uncertainty, which can improve communication with stakeholders. Industry leaders can communicate the degree of uncertainty related to future results more efficiently. This accuracy facilitates stakeholders in making well-informed decisions by accurately assessing the risks and possibilities. Our EEMD-CNN-BiLSTM-QR model contributes to sustainability efforts by delivering superior predictions. Organisations can enhance their crude oil production and distribution systems through precise energy price forecasting, controlling waste and reducing dependence on fossil fuels. This technique, in turn, contributes to reducing greenhouse gas emissions and fosters a more sustainable energy system. Finally, the EEMD-CNN-BiLSTM-QR prediction model offers a novel approach to predicting crude oil prices and a reliable hybrid model for enhancing crude oil price predictions.

## 7. Conclusion

The probability density forecast has been a challenging task in the energy market. In this paper, we have evaluated the accuracy of predictions about determined forecasting and probability density forecasts of crude oil prices achieved by integrating the time-frequency decomposition of EEMD with CNN and BiLSTM based on the QR technique. A novel short-term probability prediction model of crude oil price is conducted for two real-world datasets. The experimental results validate that applying EEMD in crude oil price models improves forecasting accuracy. The probabilistic prediction model based on DL proposed in this paper can effectively improve the robustness of the probability density forecast and achieve higher prediction accuracy than the other four selected comparative models.

This paper focuses on applying DL based on QR for the probability prediction of crude oil prices but has not considered the parameter optimisation of the DL NN. An orthogonal array tuning method recently

proposed by Taguchi (Zhang et al., 2019; Pérez-Espinoza et al., 2016), is a technique of deep NN hyperparameter optimisation that may shed new light on probability density forecasting. Moreover, this study pays attention to one-step-ahead probability forecasting of crude oil prices, which cannot provide multi-step information for decision makers. These two topics will be explored as future research directions in this domain.

## CRedit authorship contribution statement

**Yukun Bao:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Liang Shen:** Writing – original draft, Software, Methodology, Conceptualization. **Yanmei Huang:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization. **Najmul Hasan:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation. **Changrui Deng:** Resources, Investigation. **Xiaohong Zhou:** Investigation, Data curation.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data Availability

The data that has been used is confidential.

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